Refractive Gravity: An Ontological Picture and the Foundations of Mathematical Formalism

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Abstract

This monograph develops the systematic ontological architecture of **Refractive Gravity** (RefG), a unified physical framework in which gravitation, discrete matter, and the operational structure of measurability emerge from the dynamics of a single universal, pre-spatial, and pre-temporal foundation-medium. Resonant self-distinction gives rise to operational space and clock time through a network of distinguishable nodes with characteristic scale ℓ_*. Locally confined nonlinear wave configurations—termed *oscillons*—induce a static pressure deficit in the surrounding medium through Bernoulli-type dynamics ($-\Delta P \propto \rho_{\mathrm{dyn}}$), registered externally as active gravitational mass. Gravitational attraction and light deflection emerge as universal refraction across this inhomogeneous pressure landscape; along the static weak-field first post-Newtonian (1PN) branch, the biconformal mechanism yields the Parameterized Post-Newtonian parameters $\beta = \gamma = 1$ and recovers the classic predictions of General Relativity. Across astrophysical scales, the framework develops a conditional finite-core map for relativistic compact interiors; derives the MOND interpolation law and asymptotic Baryonic Tully–Fisher Relation from a macroscopic polarization potential; and reports a fixed-input comparison with 2,788 SPARC measurements, yielding an RMS scatter of 0.141 dex and a mean residual of -0.012 dex. Cosmological expansion and dark energy are interpreted through the global thermodynamic relaxation of the foundation-medium. At microscopic scales, a three-dimensional volumetric resonator and population-locked phase synchronization provide a candidate spectral map for discrete particle modes and a candidate C_3/Koide construction, while the topological invariants $Lk = Wr + Tw$ organize geometric candidate readings of spin-1/2, Dirac $g = 2$, electric charge, and color confinement within a single framework.

Contents

Preface

This monograph presents an ontological account of Refractive Gravity (RefG), unifying diverse lines of theoretical inquiry into a single, coherent physical vision. RefG is a research program under stepwise, disciplined development. The central aim of this text is to provide the reader with a deep, intuitive picture of the fundamental architecture of the Universe; the underlying computational formalism and executable verification gates are preserved in the public scientific archive on Zenodo [Lotishvili, 2026a,b].

For the current status of the hypotheses, formal proofs, and empirical boundaries discussed throughout this work, the reader is referred to the concluding section, "Current Status of the Research Program."

Introduction

Modern fundamental physics rests upon a monumental, yet internally divided, architecture. On the one hand, Albert Einstein's General Relativity describes gravitational interaction as the dynamic geometry of a smooth, four-dimensional pseudo-Riemannian spacetime; on the other, Quantum Field Theory represents matter and gauge forces as operator-valued excitations on a fixed, predominantly flat Minkowski background. Both pillars enjoy unprecedented levels of empirical confirmation, yet their internal ontological foundations remain fundamentally incompatible: gravity rejects a static background and dynamically generates the arena itself, whereas quantum mechanics, to preserve unitarity and its probabilistic structure, a priori requires a pre-determined, external temporal parameter.

At the interface of this profound disconnection, contemporary physics confronts a cluster of unresolved foundational crises:

1. **The Singularity Problem:** At the cores of black holes and at the cosmological origin, the continuous geometry of classical spacetime terminates in infinite curvature, signaling a breakdown of physical predictability.

2. **Astrophysical and Cosmological Anomalies:** Galactic rotation curves, the Radial Acceleration Relation (RAR), and the Baryonic Tully–Fisher Relation (BTFR) are conventionally accommodated by postulating dark matter halos, while the accelerated expansion of the Universe is attributed to dark energy (the cosmological constant Λ), whose theoretical vacuum expectation value exceeds observational limits by 120 orders of magnitude.

3. **Structural and Hierarchical Puzzles:** The origin of quantum measurement randomness, the physical nature of the arrow of time, the pronounced mass hierarchy among charged leptons and quarks, the unexplained values of dimensionless coupling constants (such as the fine-structure constant $\alpha \approx 1/137$), and the extreme weakness of gravity relative to the electroweak scale ($M_{\mathrm{Pl}}/m_e \sim 10^{22}$) remain arbitrary, disconnected inputs in the Standard Model.

The history of physics demonstrates that such persistent, isolated paradoxes typically indicate the absence of a deeper ontological substrate. Numerous grand unifications have sought to bridge this divide: Albert Einstein spent his final decades attempting to subsume electromagnetism and matter into an extended spacetime geometry; John Archibald Wheeler explored the nexus of geometry and information through the guiding principles of *Mass without Mass* and *It from Bit* [Wheeler, 1955, 1990]; Andrei Sakharov interpreted gravity as an elastic resistance ("induced gravity") emerging from the quantum fluctuations of the vacuum [Sakharov, 1967]; Ted Jacobson derived Einstein's field equations directly from the thermodynamic equation of state of local causal horizons [Jacobson, 1995]; Erik Verlinde formulated gravitation as an emergent holographic entropic force [Verlinde, 2011]; and Roger Penrose formalized the inescapable connection between global causal structure, gravitational collapse, and spacetime singularities [Penrose, 1965].

While these frameworks employ distinct mathematical vocabularies, they share a singular, foundational question: **Is it possible that the entire tapestry of physical phenomena—space, time,**

gravity, and the particle spectrum—is the dynamic manifestation of a single primary ontological substrate?

This monograph answers this question in the affirmative, presenting the unified conceptual and ontological framework of **Refractive Gravity** (RefG).

Operational Foundations and Three Fundamental Principles

The standard language of physics typically begins with an established measuring apparatus: smooth coordinates, a continuous time axis, and operator fields are treated as axiomatic primitives. RefG grounds its inquiry in the strict operationalism of Percy Bridgman [Bridgman, 1927]: *Before speaking of space, time, or mass, what physical process generates the local, distinguishable differences upon which rulers and clocks function?*

In 1905, Einstein dismantled the concept of absolute time by operationally defining simultaneity via clocks and light signals [Einstein, 1905]. RefG pushes this operational critique to a more primordial level: **What physical mechanism constructs the stable clock, the material particle, and the propagating wave signal upon which all such measurements fundamentally depend?**

In response to this question, RefG formulates three fundamental postulates:

1. **Monistic Ontological Foundation:** Nature possesses strictly one universal, pre-spatial, and pre-temporal foundation. Its resonant self-distinction generates a manifested medium whose phase, pressure, shear, rotational, and topological modes are operationalized as elementary fields, forces, and particles.

2. **The Oscillon Nature of Matter:** A stable massive material unit is an *oscillon*—a locally confined, self-sustaining nonlinear wave configuration of the manifested medium. Massless carrier modes are the freely propagating radiation waves of the same medium. The localized oscillatory action of an oscillon generates a persistent, long-range static pressure-deficit profile in the surrounding substrate.

3. **Gravity as Effective Refraction:** Oscillon matter establishes an inhomogeneous pressure-deficit landscape throughout the medium. This deficit alters the effective geometry, causing the wavepaths of light and matter to bend refractively. Gravitational attraction is thus revealed as the physical mechanism of spacetime curvature.

Lorentz Invariance and Historical Precedents

The Michelson–Morley experiment of 1887 detected no "aether wind" from the Earth's orbital motion, placing stringent constraints on 19th-century models of a rigid, matter-independent "luminiferous aether" [Michelson and Morley, 1887]. The RefG foundation-medium belongs to an entirely different ontological category: here, measuring rods, clocks, the observer, and propagating light are themselves collective wave excitations woven from this very medium.

When an observer and their apparatus are co-emergent wave modes of the medium, uniform translational motion corresponds to a uniform phase transformation of the wavepacket. Along the Lorentz-covariant branch, this motion leaves no dissipative wake or detectable "wind." This realization revitalizes Lord Kelvin's (William Thomson's) visionary intuition regarding "vortex atoms" [Thomson, 1867]: **Matter is not a foreign body inserted into a passive background—matter is the organized, self-sustaining motion of the medium itself.**

Seven Core Theoretical Contributions

This text establishes the ontological architecture of RefG, endowing the computational formalism archived on Zenodo with unambiguous physical meaning. While medium-based unifications have notable precedents—such as Winterberg's Planck aether model [Winterberg, 2002]—RefG places particular emphasis on the operational genesis of measurability and population-level resonant locking.

The core theoretical findings of this work are distilled into seven principal points:

1. **Population-Locked Time:** The local rate of temporal passage, $\Omega(x)$, is not an externally imposed, abstract metronome; it emerges directly from the collective nonlinear locking and resonant synchronization of the local oscillon population ($\nu_i = n_i\,\Omega(x)$). Gravitational time dilation is nothing other than the joint reconfiguration of this local resonant network in response to a shift in substrate pressure.

2. **Two-Channel Mass Accounting:** The theory rigorously decouples two physical quantities: the oscillon's internal topological state (its invariant internal energy store) and its externally read gravitational mass, which represents the volume-integrated pressure deficit induced in the surrounding medium.

3. **Biconformal p/p^2 Scaling and the Factor-of-Two Deflection:** Along the first-order biconformal branch, operational length, externally read mass, and clock rates scale linearly with substrate pressure (p), whereas the coordinate propagation of light scales quadratically (p^2), rooted in the differing geometrical nature of network nodes versus interconnecting links. In the weak-field regime, this mechanism exactly reproduces the metric predictions of General Relativity, yielding the double deflection of light and Shapiro time delay (PPN $\beta = \gamma = 1$). At macroscopic scales, this scalar-pressure channel, augmented by the non-integrable strain of topological defects and the effective field theory (EFT) framework of Fierz–Pauli and Deser [Deser, 1970], closes into the full Einstein–Hilbert action.

4. **The Koide C_3 Phase Geometry:** In the three-phase mass map of charged leptons (e, μ, τ), the mass angle $\theta = h/9$ represents the reduced coordinate of a ninth-order holonomic return cycle. On the selected candidate branch, the first non-trivial topological return gives $h = 2$ and $\theta = 2/9$, while internal vibrational-channel equipartition is associated with Koide's dimensionless modulation amplitude $A_K = \sqrt{2}$.

5. **The Nodal Origin of the Fine-Structure Constant (α):** The fine-structure constant $\alpha \approx 1/137$ is interpreted as the dimensionless impedance ratio of the substrate's electric-flux and transverse-photon response channels, remaining invariant under local pressure transformations.

6. **The Global Slowing of Time and the Bridge between Two Measures:** The local pressure–cadence law, $\Omega = \sqrt{P/P_{\text{ref}}}$, extends to cosmological scales. The historical decline of the mean pressure of the foundation-medium lowers the common cadence from epoch to epoch; on the selected background branch, $d\tau/dt = a^{-3/4}$, so internal evolutionary processes in the past ran faster when measured against present-day metric clocks. The special branch $\alpha = -1/2$ gives constant-rate vacuum relaxation in process time, while the free-α cosmological family describes nonlinear accelerated expansion in metric time. The current joint fit yields a metric age of 13.62 Gyr and a present-clock-calibrated process interval of 29.86 Gyr—two measures of one cosmological history.

7. **Two Readouts of a Single Spectral Operator—A Central Hypothesis:** The short-wavelength eigenfunctions of a unified volumetric Helmholtz/Laplace-type spatial resonator are read as microscopic particle states, whereas its long-wavelength modes are read as candidate organizers of the cosmic web and galaxy-cluster structure.

Historical Lineage and Priority Boundaries

Every component of RefG is anchored in the established heritage of theoretical physics:

- The p/p^2 biconformal scaling shares conceptual kinship with the Polarizable Vacuum (PV) frameworks of Dicke and Puthoff [Dicke, 1957, Puthoff, 2002]; in RefG, it arises from the discrete ontology of a nodal network and two-channel mass accounting rather than phenomenological dielectric permittivity.

- The three-phase geometric reading of Koide's formula extends the pioneering work of Brannen and Foot [Brannen, 2010, Foot, 1994], while the holonomic origin of 2/9 resonates with Shulga's compact family cycles [Shulga, 2026]. RefG unifies this structure with θ-blindness arguments and the C_3 tetrahedral symmetry of a 3D volumetric resonator.
- Cosmological vacuum pressure relaxation builds upon the q-theory of Klinkhamer and Volovik [Klinkhamer and Volovik, 2008], while the selection of three spatial dimensions rests on the topological knotting and stability of vortex lines [Whitrow, 1955, Berera et al., 2017].

The defining contribution of this monograph is the synthesis of these disparate threads into a single, logically connected research program: tracing the path from the operational birth of measurability and oscillon mass to galactic vortex dynamics and cosmological horizons.

The text invites the reader to follow this chain and see how a unified physical intuition unfolds into a broad panorama of contemporary fundamental physics.

1 The Ontological Foundation

1.1 A Single Foundation and Its Manifested Medium: Substance, Void, and the Dynamics of Existence

Modern fundamental physics describes the Universe as an assembly of mutually independent fields. In the Standard Model, each field is equipped with its own internal gauge symmetry, intrinsic degrees of freedom, and coupling parameters. Refractive Gravity (RefG) replaces this pluralistic ontology with a single, uncompromising monistic principle: **Nature possesses strictly one universal, pre-spatial ontological foundation.** Its resonant self-distinction generates a measurable, multi-channel medium. What we observe as distinct fundamental fields, gauge interactions, and elementary particles are the topological, phase, pressure, shear, and rotational modes of this single manifested medium. Throughout this monograph, the term "foundation-medium" denotes this self-distinguished, physically manifested state of the primary substrate.

This monistic vision possesses deep historical roots in theoretical physics. As early as 1870, William Kingdon Clifford formulated the prophetic hypothesis that matter consists of small, propagating ripples of spatial curvature [Clifford, 1876]. RefG endows this geometric intuition with a concrete physical mechanism: the self-distinguished medium supports not only geometric curvature responses, but also pressure, shear, hydrodynamic, and resonant channels.

A prominent theoretical precedent for the emergence of diverse particle and gauge sectors from a single substrate is the *string-net condensation* framework of Michael Levin and Xiao-Gang Wen [Levin and Wen, 2005]. In their model, collective many-body entanglement within a simple bosonic system naturally gives rise to both gauge bosons (photons, gluons) and fractional fermionic excitations (electrons, quarks). While the Levin–Wen model presupposes a pre-existing spatial lattice, RefG grounds this unifying capacity in the dynamic self-distinction of a single, pre-spatial foundation.

Conceptualizing the Universe as a unified continuum requires two asymptotic dynamical bounds:

1. **The Absolute Maximum of Substance:** A saturation limit characterized by maximal pressure, mass-energy potential, and operational nodal density.
2. **The Zero (Vacuum) Bound:** A limit in which interaction and physical measurability vanish entirely.

This "void" is not an abstract metaphysical nothingness, but the operational limit where physical distinguishability ceases. Objective physical reality unfolds as the bounded, nonlinear oscillatory dynamics between these two poles, wherein stable local units of the foundation—"nodes of existence"—are formed.

The working background $\varphi = 0$ is the quiescent yet physically active state situated between these two extrema. At the limit of zero distinguishability, nodal interactions and measurable traces completely vanish; $\varphi = 0$ designates the unperturbed baseline state of the manifested medium.

The ontological unity of the manifested medium is accompanied by a rich multi-channel capacity. The medium can compress (volumetric pressure), twist (topological torsion), establish vortical circulation, and support transverse wave excitations. A hydrodynamic analogy provides useful intuition: a single fluid concurrently supports isotropic hydrostatic pressure, vortex circulation, shear stress, and surface waves. Similarly, the RefG manifested medium integrates phase, pressure, shear, and topological responses, providing the dynamic substrate upon which electric charge, spin, and the particle spectrum are constructed.

The unperturbed state ($\varphi = 0$) represents the baseline vacuum regime. The foundation is omnipresent; what varies locally is its degree of excitation, the operational density of distinguishable nodes, and the coupling strength between them. A drop in pressure, or rarefaction, corresponds to an increase in the unmeasured operational intervals separating nodes.

Here, "pressure" serves as the effective state variable of energy density in RefG. The baseline $\varphi = 0$ allows deviations in both directions: $\varphi > 0$ corresponds to a compressed, high-pressure branch, while $\varphi < 0$ denotes a rarefied, deficit branch. The entire gravitational readout in RefG is constructed upon the gradient profile of this deficit branch ($\varphi < 0$). Classical Bernoulli pressure serves as the direct physical analogue: for an external observer, a local shift in φ is simultaneously registered as gravitational time dilation, coordinate retardation of light, operational length contraction, and effective gravitational mass.

1.2 From Single Existence to Multiplicity: The Emergence of Measurability and Operational Space

So long as the foundation-medium remains in a state of absolute, unperturbed uniformity, physical measurement is impossible in principle. In such a perfectly symmetric state, the concepts of "distance," "time," or "geometry" have no meaning, as there exists no second, physically distinguishable local state against which comparisons or coordinated measurements could be performed.

The physical Universe originates precisely through the self-differentiation (self-distinction) of the medium: the moment a local asymmetry—a distinguishable node, a phase twist, or a pressure deficit—arises, the foundation ceases to be a trivial, featureless background.

An interval between any two distinguishable nodes constitutes the physical embryo of space; the transmission delay of a signal between them defines time; and the localized energetic strain of differentiation represents the ontological root of mass. The externally read gravitational mass is subsequently determined by integrating the pressure-deficit profile that this strain induces in the surrounding foundation.

Physical meaning invariably requires comparison. At the ontological origin, three indivisible elements emerge:

- **Node:** A locally distinguishable, discrete state of the foundation;
- **Trace:** The energetic and phase alteration imparted by a node to the surrounding medium;
- **Ratio:** The operational capacity to compare one local state with another.

From this foundational triad unfolds the entire operational network that macroscopically manifests as space, time, and matter.

A completely undistinguished foundation possesses a single, indivisible identity. Its resonant self-distinction generates the first persistent difference, marking the birth of quantity and counting. Distinguishable manifestations are registered as separate physical units without the foundation fragmenting into disconnected parts—multiplicity consists of the stable, mutually distinct states of one fundamental identity.

The primary unit of measurement is not an isolated object, but a preserved ratio. In an undistinguished state, no second entity exists for comparison. Self-distinction yields two distinct states of the same foundation and establishes the primary link between them; here begin counting, spatial extension, and the relative ordering of events.

In information-theoretic terms, the first preserved distinction is analogous to a bit: a distinguishable trace is registered as "1", and its absence within a given relational frame as "0". Complex arrangements of these binary states generate rich macroscopic structures, just as the mutually distinct states of the foundation generate the diversity of the observable cosmos.

The concept of a single object existing at different locations at different times illustrates this logic. If the temporal labels along its worldline are removed, a single identity leaves traces across multiple locations. RefG elevates this intuition to the pre-spatial domain: persistent self-distinctions of the foundation construct the very locations that can subsequently be counted and compared. The atemporal status of the foundation and the metric age of the observable Universe thus belong to distinct conceptual levels.

This perspective is crucial for understanding the nature of the observer. Observers and their measurement instruments are themselves complex, composite wavepackets built from this nodal network. This deepens Percy Bridgman's operationalism, which asserts that physical concepts are defined solely by the operations required to measure them [Bridgman, 1927]. RefG provides this operational philosophy with an ontological substrate, investigating the physical conditions that make measurement operations possible in the first place.

The observable Universe is not matter placed inside an empty geometric container; it is the ongoing process of self-distinction within the foundation-medium. Operational space emerges along the traces of this distinguishability. Beyond these traces, the foundation may persist, but as a physical object it manifests only where it leaves an operational, measurable imprint.

The absolute separation between nodes cannot be measured directly, because no observer possesses a measuring rod external to the medium. What is measurable is strictly the operational trace of the inter-node interval. Operational space is born when the relation between at least two nodes becomes physically distinguishable. Below this threshold, variation remains an unmanifested internal state; above it, it registers as shifts in clock rates, operational sizes, mass readouts, and signal velocities.

RefG posits a dynamic lower bound of minimal operational distinguishability, denoted by ℓ_*:

- At scales smaller than ℓ_*, no independent measurable ratio can form (information is undistinguishable);
- At the scale ℓ_*, the first stable, physically distinguishable record emerges;
- At scales larger than ℓ_*, inter-node intervals are operationalized as variations in pressure, clock rates, and propagation velocities.

The physical rationale for this bound is that an infinitely fine record would lack dynamical stability. While the Planck length $\ell_{\mathrm{Pl}} = \sqrt{\hbar G/c^3}$ is a dimensional scale constructed from fundamental constants, ℓ_* represents the dynamical threshold for stable record formation. Their potential coincidence depends on the microdynamics of the foundation.

Consequently, the ontological origin of the Universe is not a pre-existing manifold of geometric points. Prior to the emergence of inter-node intervals, there are neither distinct points nor spatial extension. Such a state cannot even be termed a "single point," as a point is a concept belonging to an already realized geometry; it is an undistinguished primary ground state. Multiplicity and geometry emerge only when an asymmetry generates a stable trace, rendering inter-node ratios operationally readable.

Here lies the physical essence of gravity: when local coupling between nodes weakens (pressure falls), this state is simultaneously registered by an external observer as clock retardation, contraction of operational scale, and signal delay. This coordinated triad of physical responses is precisely what macroscopic observers perceive as a "gravitational field."

1.3 Nodal Transmission: Waves, Time, and Self-Calibration

The self-distinction of the foundation generates an interconnected network of nodes. Each node represents a functional, distinguishable state of the foundation: it stores the trace of a physical change and relays it to adjacent nodes. Spatial proximity and operational distance emerge from the persistent topology of these linkages.

A wave in this network is the sequential transmission of state and phase. Light is a freely propagating transverse phase trace, whereas an oscillon is a locally confined, self-sustaining configuration of the same transmission mechanism. When matter moves, this confined wavepacket is transferred from node to node, reconstructing its phase integrity at each successive location. Light and matter are thus the propagating and localized modes of a single medium.

Every transmission step establishes a relative ordering of events. Periodic phase relations structure this sequence into a pre-clock ordering, while comparisons between stable oscillon cycles yield clock time. The local rate of time is the collective rhythm of the network: the speed at which the medium relays state information and the rate at which an oscillon completes its internal cycle.

Thermal insulation in double-pane windows provides a helpful analogy. Rarefying the intermediate gas reduces molecular collisions and slows the rate of heat conduction. RefG extends this intuition to the nodal network, linking the transmission rate directly to nodal density and coupling strength.

The observer's body, clocks, rulers, light, and matter are all modes of this same medium. In a deficit region, nodal density, transmission rates, oscillon cycles, and operational sizes undergo a coordinated shift. Local measuring devices reconfigure synchronously with their environment; consequently, an observer using local standards measures normal clock rates, standard dimensions, and the invariant local speed of light (c). Comparing processes that have traversed different environments, however, reveals objective differences in accumulated time, operational size, and mass.

Along the selected Lorentz-covariant low-energy branch, this transmission is isotropic, non-dissipative, and strictly respects Lorentz symmetry: the local speed of light remains constant, and uniform motion leaves no frictional wake.

1.4 Local Collective Cadence and Resonant Locking

The common cadence of local temporal passage is not an abstract, external metronome imposed upon the foundation-medium. It emerges dynamically the moment stable oscillons (the wave configurations of matter) integrate into a unified local resonant network, locking their intrinsic frequencies into fixed, stable ratios. In such a collective state, the rate of time is generated endogenously through the nonlinear mutual locking and synchronization of the local network (analogous to Kuramoto-type synchronization).

From an ontological perspective, the foundation is neither a tone nor a pre-assembled clock. The manifested medium produced by its self-distinction acts as an acoustic instrument—a three-dimensional vibrating volumetric resonator that imposes boundary conditions, supplies the energetic reservoir, and determines the allowed normal-mode spectrum.

The physics of a bell offers a vivid illustration: the acoustic signature of a bell is not determined by isolated, arbitrary frequencies, but by the normal modes of a continuous elastic body resonating in collective harmony. Similarly, the frequencies of elementary particles are coupled through mutual balance within the energetic reservoir of the foundation. The universal foundation sets the allowed modal arena, while particle modes mutually equilibrate across this arena to form a stable, resonantly locked system.

Denoting the local collective cadence by $\Omega(x)$, the frequency spectrum of stable oscillons is expressed as:

$$\nu_i(x) = n_i \, \Omega(x), \tag{1}$$

where n_i is a fixed, dimensionless structural factor characteristic of a specific particle mode. In general, n_i is non-integer, meaning that particle mass and frequency spectra do not reduce to simple one-dimensional overtones; rather, they reflect the complex, slightly anharmonic eigenspectrum of a 3D volumetric resonator (as detailed in the C_3 symmetry of charged leptons).

The relation $\nu_i(x) = n_i\,\Omega(x)$ preserves individual mode identities while locking every mode to the shared local cadence $\Omega(x)$. Consequently, the frequency ratios:

$$\frac{\nu_i(x)}{\nu_j(x)} = \frac{n_i}{n_j} = \text{const} \tag{2}$$

remain strictly invariant under resonant locking. Any oscillatory rhythm incompatible with the collective network loses phase coherence and cannot persist as a stable particle mode.

In a region of low pressure (gravitational deficit), the collective cadence $\Omega(x)$ of the entire local network decreases. In a dense, high-pressure substrate, inter-node coupling is robust and $\Omega(x)$ has a higher value; in a rarefied substrate, coupling weakens and $\Omega(x)$ slows down:

- The retardation of inter-node signal transmission,
- The reduction in substrate pressure-energy density, and
- The reduction of the collective cadence $\Omega(x)$

represent three distinct operational readouts of a single ontological reality; within the formal mathematical chain, each enters exactly once.

When $\Omega(x)$ shifts locally, all local clocks, atomic transitions, and particle processes scale synchronously. Consequently, a local observer cannot detect any variation in their own rate of temporal passage. Under this collective rescaling, all dimensionless ratios—including the fine-structure constant α and mass ratios m_i/m_j—remain strictly invariant, reflecting the internal geometric-topological proportions of the network rather than its absolute rate.

The historical precursor to this relational approach is Ernst Mach's principle [Mach, 1883]. RefG reformulates Mach's principle in a localized, dynamical framework: the common rate of time is governed not by the instantaneous configuration of distant cosmic masses, but directly by the collective state of the local resonant network. In this framework, relativistic invariance and local causality emerge as intrinsic features of the local network.

In the RefG framework, the fine-structure constant $\alpha \approx 1/137$ is interpreted as the dimensionless impedance ratio of the substrate's electric-flux and transverse-photon response channels, remaining invariant under local shifts in substrate pressure.

Now consider a wavepacket propagating across differing pressure regimes of the foundation (for example, from a deep gravitational well to an asymptotic observer). Its observed frequency is computed via the effective spacetime metric. In the ontological language of RefG, the local rates of source and receiver are compared through their phases against their respective local cadences, while cosmological redshift $1 + z = a_0/a_{\text{em}}$ is determined by standard metric relations. The signal does not carry a "frozen," absolute frequency across space; its phase evolution is continuously referenced to the local $\Omega(x)$ cadence of each region it traverses. A compatible signal locks into the new resonant network, whereas an incompatible one undergoes dephasing or scattering, redistributing its energy into the pressure-energy channels of the substrate.

In summary, the rate of time in RefG is the ontological state of the local resonant network. When substrate pressure shifts, the network undergoes a collective rescaling: all compatible processes scale synchronously, leaving dimensionless ratios strictly invariant. An independent, abstract universal time possesses neither operational utility nor physical meaning: every physical measurement is executed strictly from within the network itself.

This principle establishes a rigorous empirical falsification criterion: **If, under a local pressure shift (such as within an intense gravitational field), any dimensionless ratio (such as $\Delta\alpha/\alpha$) varies in a manner inconsistent with the collective scaling of $\Omega(x)$, the postulate of a population-locked cadence is decisively refuted.**

1.5 Oscillons, Pressure Deficits, and the Origin of Mass

What traditional physics designates as an "elementary particle"—whether an electron, proton, or quark—is recognized in RefG as an **oscillon**. An oscillon is a locally confined, self-sustaining, nonlinear topological wavepacket (a soliton-like state) supported by the foundation-medium. It is not an isolated, solid corpuscle distinct from the background; rather, it is a highly organized, coherent oscillatory state of the medium itself.

This concept continues the profound lineage of Lord Kelvin's (William Thomson's) "vortex atoms" [Thomson, 1867]. While 19th-century mechanical aether models were superseded, Kelvin's core insight—that particles are dynamic, self-confined topological modes of a continuous medium—remains a foundational concept that oscillon physics formalizes within modern nonlinear field theory and topology.

When an oscillon oscillates intensely, rotates, or confines energy within its localized geometry, the static pressure of the surrounding foundation-medium decreases. This behavior is illuminated by Daniel Bernoulli's hydrodynamic principle [Bernoulli, 1738]: **An increase in internal dynamic intensity (velocity, oscillation amplitude) induces a corresponding drop in local static pressure.** On the weak, static gravitational branch of RefG, this hydrodynamic analogue corresponds to the schematic pressure-energy relation:

$$\Delta P \propto -\rho_{\mathrm{dyn}}. \tag{3}$$

Through its continuous oscillatory action, an oscillon imprints a stable static pressure-deficit gradient upon the surrounding foundation. Within this framework, this pressure deficit is nothing other than the **direct external profile of gravitational mass**: integrating this deficit over all space yields the total gravitational mass of the system. The more intense the oscillon's internal dynamics, the deeper the induced pressure deficit and the stronger the resulting refractive gradient. Simultaneously, the oscillon's internal topological energy store is accounted for as a separate, decoupled physical channel.

The core of an oscillon is localized, but its oscillatory trace extends throughout the connected medium. The component phase-locked to the source establishes a static gravitational profile, requiring zero net outward energy flux. When the source undergoes rapid dynamic reconfiguration, the coherent oscillatory component released from this channel propagates outward as a gravitational wave, carrying energy and momentum.

The local depth of the deficit and the asymptotically measured total mass of a system are distinct physical quantities. The former characterizes central gradient and compactness; the latter integrates the active core, substrate dressing, binding energy, and radiated flux into a single asymptotic balance. Consequently, following a binary merger, the central deficit deepens while the total externally read mass decreases by the exact amount of radiated gravitational-wave energy.

This perspective shares structural parallels with John Archibald Wheeler's concept of *Mass without Mass*, in which a "geon" represents a massive entity bound entirely by the self-energy of its gravitational and electromagnetic fields [Wheeler, 1955]. While classical Wheeler geons are unstable to radiative decay, RefG stabilizes oscillons through topological charge conservation, resonant phase locking, and the topological knot invariant $Lk = Wr + Tw$.

Consequently, the term "effective refractive mass" (m_{eff}) designates the externally read gravitational mass of an individual oscillon within a specified background medium. Under adiabatic background variations, the oscillon's topological structure, internal energy, and phase ratios preserve their identity, while its operational scale and externally read mass adapt to the new substrate state. Rapid reconfiguration represents an independent dynamical process that redistributes energy into radiation channels. The total asymptotic mass of an extended system requires a volume integration and is not simply the linear sum of uncoupled single-oscillon m_{eff} values. Through this decoupling, the origin of mass is revealed as the establishment of the medium's refractive response.

At this foundational level, **gravitational attraction is the refractive bending of one oscillon's wavepacket within the pressure deficit generated by another.** A relaxed string provides an intuitive picture of a gravitational well: locally diminished pressure-energy weakens the effective wave propagation stiffness of the substrate. The deficit profile defines the effective geometry, and any matter wavepacket bends along geodesics of minimal optical path length according to the eikonal equation.

This defines the central dynamical reciprocity proposed by RefG: **An oscillon creates a pressure deficit through its internal oscillation, while this deficit determines its effective mass, operational scale, and trajectory through the medium.** Matter dictates the pressure state of the medium, and the altered medium dictates the refractive geometry of matter.

2 Gravity as Refraction

2.1 Three Simultaneous Operational Shifts

As established in the preceding chapter, the inter-node separation within the foundation-medium is not directly accessible to external measurement; an observer can access only its operational, physical traces. Around an oscillon, this trace manifests as a localized depression in the pressure-energy state of the substrate: distinguishable nodes are rarefied, and the informational and wave coupling between them is attenuated. For an external, macroscopic observer, this single ontological state manifests as three simultaneous, coordinated operational shifts:

1. **Deceleration of the Coordinate Speed of Light:** In a pressure-deficit zone, nodes are rarefied and the rate of inter-node transmission drops. Consequently, the propagation of light and all other massless wave signals is retarded in coordinate description. The locally measured speed of light remains strictly invariant (c) for a local observer; what changes is the effective propagation rate evaluated in the coordinate language of a distant observer.

2. **Contraction of Oscillon Operational Size:** While the oscillon's intrinsic topological form (its nodal winding configuration) remains invariant, it is populated by a reduced volumetric density of interacting nodes within the rarefied substrate. In external observational description, this is registered directly as a contracted operational size, manifesting as a modified effective mass readout.

3. **Retardation of the Local Resonant Clock Cadence:** Real physical clocks are composed of oscillon systems (atoms, molecules) whose internal tick rate depends directly on local nodal coupling. In a rarefied environment, this coupling weakens, decelerating the clock rate in coordinate time relative to an asymptotic observer. Locally, the clock continues to measure its own cadence normally and invariantly.

A physical analogy is provided by double-pane vacuum insulation: evacuating the intermediate space reduces molecular collision frequency and suppresses the rate of thermal transport. In the RefG framework, this corresponds to the relation between nodal coupling strength and coordinate clock speed.

These three shifts are distinct operational readouts of a single underlying pressure deficit. Crucially, they scale with substrate pressure according to different powers, establishing the foundation of biconformal geometry.

Nodal rarefaction does not generate a "secondary space." The inter-node interval is not an independent physical substance; macroscopic "void" is operationalized entirely through the language of pressure variables. The sole objective manifestation of substrate rarefaction is the localized reduction of its pressure-energy state.

2.2 Quadratic Coupling and the Factor of Two

These three operational shifts do not scale with substrate pressure under a uniform power law. In the ontological architecture of RefG:

- **Operational size and single-oscillon effective mass** belong to the volumetric nodal capacity channel (extensive quantities);

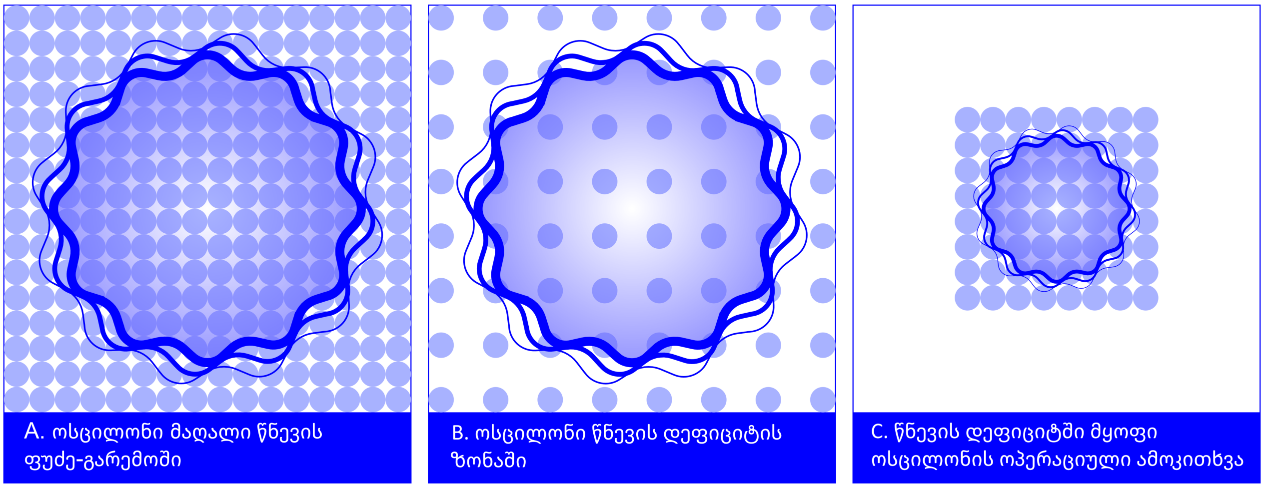

Figure 1: **Operational readout of an oscillon within a pressure-deficit zone. (A)** In an unperturbed, high-pressure foundation-medium, an oscillon resides in a dense nodal network; inter-node coupling is robust and local processes proceed at maximal cadence. **(B)** Within a pressure-deficit zone, substrate nodes are rarefied; operational inter-node distance increases, transmission rates decrease, and local temporal passage decelerates. **(C)** In external readout, an oscillon in a pressure deficit is registered with a slowed clock rate, contracted operational size, and shifted effective mass, while the internal separation does not enter as an independent direct observable.

- **The coordinate speed of light** belongs to the inter-node coupling strength and transmission delay channel (intensive quantity).

Denoting the dimensionless pressure factor by p, this biconformal scaling is formalized as:

$$\frac{L_{\text{oper}}}{L_0} = \frac{m_{\text{eff}}}{m_0} = \frac{\Omega}{\Omega_0} = p, \qquad \frac{c_{\text{coord}}}{c_0} = p^2, \qquad p = e^{\varphi/2}. \tag{4}$$

A similar power-law scaling appears historically in the Polarizable Vacuum (PV) models of Robert Dicke and Harold Puthoff [Dicke, 1957, Puthoff, 2002]. In RefG, these powers are assigned to distinct nodal-capacity and link-transmission channels rather than introduced through a phenomenological dielectric ansatz.

Formally, the external macroscopic readout of a single oscillon couples to the substrate pressure deficit through an exponential half-factor:

$$m_{\text{eff}} = m_0 e^{\varphi/2}. \tag{5}$$

On the deficit branch, $\varphi < 0$, confining the pressure factor to $0 < p < 1$. This relation describes the local external readout of an individual oscillon embedded within an inhomogeneous background: as φ deepens (pressure drops), its operational size and m_{eff} contract. Simultaneously, the total gravitational charge of the extended source generating the deficit is determined via a separate volumetric integration. This scaling follows directly from the theory's first-order biconformal branch.

Here, four physical quantities must be rigorously distinguished:

1. **Oscillon internal state and proper energy** (M_{proper});
2. **Single-oscillon effective refractive mass in a given background** (m_{eff});
3. **Active source profile of the induced pressure deficit** (M_{active});
4. **Total asymptotic gravitational charge of the system** (ADM/Komar mass: Arnowitt et al. 1962, Komar 1959).

These four quantities organize into two nested accounting channels: the first distinguishes proper internal content from background-read mass; the second relates the externally read mass to the active core and substrate dressing. At the asymptotic boundary, these contributions combine into a single conserved charge, ensuring that every deficit trace is counted exactly once.

This self-consistent dynamic is captured by an intuitive physical analogy: **Imagine a crystal of colored ice dissolving in pure water.** The crystal is a locally frozen, ordered phase of the same water—just as an oscillon is a localized resonant locking of the universal foundation-medium. As the crystal melts, its dye disperses throughout the water, coloring the medium while the crystal itself shrinks in size. Energy is conserved: like the dissolved dye, localized binding energy transitions into an extended substrate pressure-energy state.

The more intensely the foundation is "colored" by this trace (i.e., the deeper the pressure deficit), the further local pressure drops, altering the externally read mass, operational size, clock cadence, and coordinate speed of light. This is a self-regulating relaxation process: the more trace an oscillon imparts to the medium, the lower its capacity for further emission. This negative feedback provides a natural mechanism for deficit saturation, with its precise boundary governed by full nonlinear dynamics.

This p/p^2 scaling is formalized through the biconformal metric:

$$ds^2 = e^{\varphi}c^2dt^2 - e^{-\varphi}d\boldsymbol{\sigma}^2, \qquad \varphi = -\frac{2U}{c^2}, \tag{6}$$

where $U > 0$ is the gravitational potential of the selected exponential branch, coinciding at leading order with the Newtonian potential. In the weak-field limit ($U/c^2 \ll 1$), the metric components expand in a Taylor series:

$$g_{00} = e^{\varphi} \approx 1 - \frac{2U}{c^2} + \frac{2U^2}{c^4} + \cdots, \qquad g_{ij} = -e^{-\varphi}\delta_{ij} \approx -\left(1 + \frac{2U}{c^2} + \cdots\right)\delta_{ij}. \tag{7}$$

On the selected static, spherically symmetric first post-Newtonian (1PN) branch, expanding this metric yields the Parameterized Post-Newtonian (PPN) coefficients:

$$\beta = 1, \qquad \gamma = 1. \tag{8}$$

A complete clock cycle is operationalized by the ratio of operational size to coordinate velocity ($\Delta t \sim L_{\text{oper}}/c_{\text{coord}}$). If a pressure drop halves the operational size ($p = 1/2$) while reducing coordinate light speed by a factor of four ($p^2 = 1/4$), the period doubles—the clock runs precisely twice as slow.

Along this biconformal branch, the 1PN effects controlled by $\beta = \gamma = 1$ identically recover the classical predictions of General Relativity: the gravitational deflection of light, Shapiro time delay, gravitational redshift, and planetary perihelion precession [Bertotti et al., 2003, Will, 2014]. In particular, the Cassini radio-link experiment measured $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$.

Historically, the additional **factor of two** arising from the spatial metric component g_{ij} was the decisive contribution that elevated Einstein's 1911 half-deflection (derived solely from temporal dilation) to the full geometric result of 1915–1916 [Einstein, 1916, Shapiro, 1964]. In RefG, this factor of two emerges naturally from the differing ontological scaling of nodal volumetric density (p) and link transmission delay (p^2).

2.3 Refractive Attraction, Inertia, and the Absence of Friction

An oscillon, as an extended wave configuration, exists within the inhomogeneous gradient of substrate pressure and coordinate light speed. On the side where the medium is denser, wave phase propagates rapidly; on the rarefied (deficit) side, propagation is comparatively retarded. By virtue of the Huygens–Fresnel principle and the eikonal equation, the wavefront bends toward the region of lower pressure and reduced coordinate velocity—undergoing **refraction**. In the static weak-field limit, the pressure deficit generated by a massive body constructs an effective optical-geometric medium, and the refractive trajectory of a matter wavepacket within this medium is perceived macroscopically as gravitational attraction.

In RefG, the classic phenomenon of Newton's falling apple is understood as follows: the oscillon wavepackets of the atoms comprising the apple bend refractively toward the region of lower pressure and slower coordinate propagation established by the Earth's mass. Familiar gravitational attraction is not an occult action-at-a-distance force—it is the direct, local refractive response of matter waves to an inhomogeneous medium.

Formally, the gravitational source consists of the projected pressure deficit and non-integrable strain in the foundation-medium. Oscillons, acting as topological defects of the substrate, introduce residual plastic strain into the continuum. In the geometric theory of elasticity, defect density is measured by the strain incompatibility tensor, which in four dimensions matches the algebraic structure of the Einstein tensor $G_{\mu\nu}$, while freely propagating transverse shear waves preserve volume and naturally manifest as transverse-traceless (TT) polarizations. A gravitational well is a 3D zone of depressed substrate pressure and strain, wherein the total mass of a bound gravitational system is strictly less than the sum of its isolated constituents by the mass equivalent of the binding energy ($\Delta M = B/c^2$).

The gravitational-wave event GW150914 provides a sharp observational benchmark for this energy balance: approximately three solar masses ($3\,M_\odot$) separating the initial binary mass from the final remnant were carried away as gravitational-wave energy [Abbott et al., 2016]. In RefG, two oscillatory sources merge into a single, deeper deficit profile, while the energy of their nonlinear reconfiguration radiates outward as transverse-traceless (TT) tensor waves. The quantitative benchmark of this framework is the simultaneous derivation of remnant mass, radiated energy, and complete waveform from a single initial state.

On the compact branch required for regular interior cores, the projected deficit source is registered as an effective violation of the Null Energy Condition (NEC); in RefG, this belongs to the elastic confinement channel of the foundation-medium.

RefG associates **inertia** with the dynamic deformation of an oscillon's deficit profile during acceleration. When an oscillon is accelerated by an external force, its pressure profile develops a fore-aft asymmetry along the direction of motion; this gradient asymmetry generates a reactive resistance from the medium opposing the acceleration (the inertial force). Under deceleration, the profile reverses symmetrically.

Along a Lorentz-covariant, localized finite-energy branch satisfying the Laue condition ($\int T^i{}_i d^3x = 0$), the inertial coefficient is determined by the total Noether energy of the system [Noether, 1918, von Laue, 1911]. Coupling this same energy as the universal source of gravity yields the equivalence $m_i = m_g$ at leading order. The French satellite mission MICROSCOPE confirms the universality of free fall to a precision of 10^{-15} [Touboul et al., 2022].

For an oscillon in uniform, rectilinear motion, the wave pattern is transferred symmetrically across substrate nodes, leaving no dissipative wake or net energy loss. RefG identifies **frictionless inertial motion** with this coherent phase-relay regime: motion is the wave-like relay of form. Its limiting velocity is set by the maximum rate of phase transmission between nodes, from which the fundamental speed of light (c) emerges.

2.4 Local Unobservability and the Principle of Relativity

A local observer residing within a gravitational field reads clock rates and spatial dimensions as standard and unperturbed, because the observer's clocks, measuring rods, and physical body are themselves constructed from the same foundation-medium. When substrate pressure shifts, the entire local physical system reconfigures synchronously. Relative to internal atomic transitions, slow variations in the gravitational field are strictly adiabatic: they preserve the vibrational mode and induce no spurious quantum transitions. Under this collective scaling, all dimensionless ratios—including the fine-structure constant α—remain absolutely invariant.

This ontological picture is constrained by comprehensive experimental bounds:

- **Gravitational Redshift:** Confirmed with exceptional precision from the classic Pound–Rebka experiment [Pound and Rebka, 1960] to the hydrogen maser on Gravity Probe A [Vessot et al., 1980] and recent eccentric-orbit tests with Galileo satellites [Delva et al., 2018, Herrmann et al., 2018], verifying the local dependence of clock rates on the gravitational potential.

- **Invariance of Dimensionless Ratios:** High-resolution quasar absorption spectra and laboratory optical atomic clocks constrain cosmological drifts in the fine-structure constant to $|\Delta\alpha/\alpha| \lesssim 10^{-6}$ [Kotuš et al., 2017]. RefG interprets this stability as consistent with local population-locked self-calibration.

The null result of the Michelson–Morley experiment reflects the medium's **common-mode self-calibration**, whereas gravitational-wave detections by LIGO, Virgo, and KAGRA represent a **differential tensor response**:

- When the foundation-medium shifts uniformly and isotropically across a local domain, measuring rods, clocks, and optical propagation reconfigure synchronously in common mode, producing common-mode self-calibration in local measurements (modern optical cavities rule out spatial anisotropy in the speed of light to $\Delta c/c \lesssim 10^{-17}$; Herrmann et al. 2009).

- A gravitational wave, by contrast, is a time-dependent, quadrupolar differential strain $(h_+, h_\times)$: it stretches space along one transverse axis while compressing it along the perpendicular axis. This differential deformation breaks common-mode cancellation, imprinting a measurable phase shift in interferometer arms.

FitzGerald–Lorentz length contraction [FitzGerald, 1889, Lorentz, 1892] is the historical precursor to this mechanism. RefG attributes this contraction to the fact that measuring instruments are themselves wave configurations of the medium.

In the local Minkowski limit, tensor waves propagate at the speed of light ($v_{\rm gw} = c$). The simultaneous detection of gravitational waves and a gamma-ray burst from the binary neutron star merger GW170817 tightly constrains the fractional difference between gravitational and electromagnetic propagation speeds:

$$-3 \times 10^{-15} \leq \frac{v_{\rm gw} - c}{c} \leq +7 \times 10^{-16} \tag{9}$$

[Abbott et al., 2017]. Concurrently, the orbital decay of relativistic binary pulsars (PSR B1913+16, PSR J0737-3039A/B) confirms general-relativistic quadrupole radiation damping to an accuracy of 10^{-3}–10^{-4} [Weisberg and Huang, 2016, Kramer et al., 2021], while bounds on scalar breathing modes from the GWTC-3 catalog tightly constrain scalar wave channels [LIGO Scientific Collaboration et al., 2022].

The most definitive operational invariant of this framework is the proper time difference accumulated by two clocks traversing distinct gravitational and kinematic worldlines:

$$\Delta\tau = \int \sqrt{g_{\mu\nu} dx^\mu dx^\nu}. \tag{10}$$

The elapsed time displayed on reunited clock faces is an objective, absolute physical record of distinct pressure histories, whereas instantaneously measured local rates and sizes remain relational readouts within the common substrate.

2.5 Historical Context and the Role of RefG

The refractive, optical-medium reading of gravity possesses an illustrious heritage. In 1911–1912, prior to completing the geometric formulation of General Relativity, Albert Einstein related the coordinate speed of light to the gravitational potential via:

$$c(\Phi) = c_0 \left(1 + \frac{\Phi}{c^2}\right) \tag{11}$$

[Einstein, 1911, 1912]. Medium optics has even earlier roots: in 1851, Hippolyte Fizeau demonstrated that moving water partially drags light (governed by Fresnel's dragging coefficient; Fizeau 1851), and in 1923, Walter Gordon proved that electromagnetic wave propagation in a moving dielectric medium obeys the null geodesics of an effective Riemannian metric [Gordon, 1923], establishing the first formal bridge between continuum optics and spacetime geometry.

In 1957, Robert Dicke formulated gravitation through a variable vacuum refractive index [Dicke, 1957]. The modern discipline of **analogue gravity** was inaugurated by William Unruh in 1981, demonstrating that sound waves in transsonic fluid flows propagate according to an effective acoustic metric exhibiting an event horizon [Unruh, 1981].

Grigory Volovik's condensed-matter program [Volovik, 2003] and the extensive review by Barceló, Liberati, and Visser [Barceló et al., 2011] established how local Lorentz invariance and curved effective geometries emerge from the microscopic dynamics of quantum condensates (such as superfluid Helium-3 and Helium-4). A notable milestone was achieved in 2019, when Jeff Steinhauer's group experimentally observed the acoustic analogue of thermal Hawking radiation in a Rubidium Bose–Einstein condensate [Muñoz de Nova et al., 2019].

Parallel to the medium-optics approach stands the **thermodynamic and entropic** school of gravity:

- Andrei Sakharov interpreted gravity as an elastic vacuum phenomenon induced by quantum fluctuations [Sakharov, 1967];
- Ted Jacobson demonstrated that Einstein's field equations arise directly as an equation of state from the thermodynamic entropy balance ($\delta Q = TdS$) across local Rindler horizons [Jacobson, 1995];
- Erik Verlinde formulated gravitational attraction as an entropic force driven by holographic information gradients [Verlinde, 2011].

These frameworks share with RefG the core conviction that spacetime geometry is emergent; in RefG, however, the effective geometry is generated not by abstract holographic screens, but by a concrete, pressure-refractive foundation-medium.

Refractive Gravity (RefG) introduces four fundamental ontological contributions to this historical continuum:

1. **Monistic Ontological Unity:** The foundation-medium is not merely a passive background for light; it generates matter itself—topological oscillons, measuring instruments, and observers. This moves beyond a passive-medium analogy and defines an independent, testable research framework.
2. **Local Population-Locked Cadence ($\Omega(x)$):** Nonlinear synchronization across the oscillon population renders local relativistic unobservability and the invariance of dimensionless constants an endogenous dynamical consequence of the system.
3. **The Universal Pressure-Deficit Channel:** RefG employs the pressure deficit ($\Delta P \sim -\rho_{\mathrm{dyn}}$) as a unified language of source and readout, linking gravitational mass, time dilation, the factor-of-two light deflection, and tensor wave responses into a single energy balance.
4. **Topological Defect Strain and EFT Architecture:** The non-integrable strain imprinted by oscillon defects (strain incompatibility; Kleinert 2008) organizes at the macroscopic effective field theory (EFT) level into the Einstein–Hilbert action through Lovelock's uniqueness result and Deser's self-coupling construction [Lovelock, 1971, Deser, 1970], with the Einstein tensor $G_{\mu\nu}$ reflecting substrate defect density.

In this historical landscape, RefG unifies optical refraction, topological solitons, and thermodynamic equilibrium into an integrated research program, with every link subject to formal mathematical and observational verification.

3 Three Scales and Cosmological Horizons

3.1 The Local Scale: Compact Objects, Internal State, and External Readout

The universal pressure chain constructed throughout the preceding chapters manifests across three distinct astrophysical regimes:

1. **On the Local Scale:** As the localized pressure-deficit profile surrounding oscillons and compact relativistic bodies;
2. **On the Galactic Scale:** As the coherent, large-scale vortical flow and polarization of the foundation-medium;
3. **On the Cosmological Scale:** As the global thermodynamic relaxation of the foundation-medium and the co-existence of two temporal languages.

These three phenomena are distinct dynamical regimes of a single foundation. We begin at the local scale, where this mechanism reaches its most extreme expression: compact relativistic objects.

In extreme gravitational regimes, an object's **internal state** (M_{proper}) and its **externally read active gravitational mass** (M_{external}) decouple fundamentally. The RefG compact map strictly distinguishes the invariant internal energy reserve of accumulated matter from its externally read gravitational charge. An external observer measures the system through its pressure-deficit profile, clock dilation, and operational scale contraction, whereas the interior geometry, horizon formation, and global causal structure are determined by the complete nonlinear field equations.

Two-channel accounting separates an object's local internal content from its asymptotic gravitational readout. The internal channel tracks local matter states, conserved topological charges, and proper volume; the external channel registers the deficit profile, clock retardation, operational dimensions, and total asymptotic mass. Gravitational compression sharply differentiates these two readouts without severing their common physical origin.

RefG attributes the mass defect of merging compact binaries to nonlinear reconfigurations in binding energy and the deficit profile. The quantitative connection to outgoing gravitational-wave flux is governed by the complete dynamical solution, benchmarked observationally by the binary black hole merger GW150914 [Abbott et al., 2016].

The biconformal metric governs the disparity between internal proper dimensions and external coordinate measures. On the compact branch of RefG, this divergence can become extreme: externally, such an object may appear ultra-compact, dark, and virtually "frozen," while its internal proper volume remains vast. In both descriptions, the locally measured speed of light is invariant (c); what differs is the temporal and spatial readout constructed by a distant observer. The "frozen star" appearance is an external observational feature, while the historical intuition of a "dark star" dates back to John Michell in 1784 [Michell, 1784].

RefG interprets a **singularity** as a boundary of external readout: under intense deficit conditions, the externally read value of the internal scale approaches zero, while the local proper scale halts at a finite saturation limit, preventing internal collapse to zero volume. A unified description requires that local energy density and spacetime curvature remain strictly finite.

Roger Penrose's singularity theorem [Penrose, 1965] relies on specific energy conditions, notably the Null Energy Condition (NEC: $T_{\mu\nu}k^{\mu}k^{\nu} \geq 0$). On the regular-core branch of RefG, the effective pressure source violates the classical NEC within the core zone via substrate elastic tension and confinement stresses. Because this energy condition is breached, Penrose's theorem does not enforce a geometric singularity on this branch, admitting a **regular, non-singular interior core**.

Regular compact objects represent an active frontier in relativistic astrophysics. A prominent example is the Mazur–Mottola "gravastar"—a vacuum condensate object devoid of a singular core [Mazur and Mottola, 2004]. Such alternatives can be empirically distinguished from classical Kerr black holes via their Event Horizon Telescope (EHT) shadow morphology [Event Horizon Telescope Collaboration, 2019], quasinormal mode ringing, and late-time gravitational-wave "echoes" [Cardoso and Pani, 2019]. RefG shares structural kinship with this family, anchoring its regular interior in the pressure-energy balance of the universal foundation-medium.

3.2 The Galactic Scale: Vortical Flow and MOND

Observational Reality. In disk galaxies, rotation curves at large radii do not follow the Keplerian decline expected from visible baryonic matter alone: they exhibit rich morphological variety, frequently flattening or declining far more gently than Newtonian predictions. Concurrently, baryonic mass and asymptotic rotation speed obey a strict empirical law—the Baryonic Tully–Fisher Relation ($M_b \propto v_f^4$; Tully and Fisher 1977, Lelli et al. 2016). This observational program began with Vera Rubin and Kent Ford's measurement of the Andromeda galaxy [Rubin and Ford, 1970]. In the modern **Radial Acceleration Relation (RAR)**, 2,693 independent data points from 153 galaxies collapse onto a single universal curve [McGaugh et al., 2016].

The standard ΛCDM model accommodates these observations by postulating spherical dark matter halos. Modified Newtonian Dynamics (MOND), by contrast, introduces a universal acceleration threshold:

$$a_0 \approx 1.2 \times 10^{-10}\ \mathrm{m/s}^2, \tag{12}$$

below which gravitational acceleration transitions from Newtonian a_N to the deep-MOND regime $\sqrt{a_N a_0}$ [Milgrom, 1983]. Relativistic completions (such as TeVeS and Skordis–Złośnik models) incorporate additional dynamical vector and scalar fields [Bekenstein, 2004, Skordis and Złośnik, 2021].

The RefG Vortical Mechanism and Substrate Polarization. At galactic scales, RefG attributes the additional deficit to a large-scale **vortical flow and polarization** of the foundation-medium. An ordered rotational regime deepens the central pressure deficit, while baryonic oscillons and the background flow form a single, shared refractive profile.

At the macroscopic level, this response is described by a Ginzburg–Landau-type local effective potential:

$$W(g_v, g_N) = \frac{1}{3}g_v^3 + \frac{1}{2}g_N g_v^2 - a_0 g_N g_v, \tag{13}$$

where g_v is the foundation-medium response field, g_N is the baryonic Newtonian acceleration, and $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 is the universal transition acceleration scale. The variational equilibrium condition, $\partial W/\partial g_v = 0$, yields:

$$g \equiv g_N + g_v, \qquad g_N = \frac{g^2}{g + a_0}. \tag{14}$$

In the deep asymptotic regime ($g \ll a_0$), this relation approaches $g^2 \simeq a_0 g_N$, yielding for circular orbits the Baryonic Tully–Fisher Relation ($v^4 = GM_b a_0$) without individual galaxy tuning parameters.

Observational Check on SPARC Data. Distinct from the published 153-galaxy, 2,693-point RAR sample, the cuts $\Delta V/V < 10\%$ and $V_{\mathrm{obs}} > 20$ km/s select 2,788 points from 163 SPARC galaxies. With fixed a_0 and fixed mass-to-light ratios ($\Upsilon_{\mathrm{disk}} = 0.5$, $\Upsilon_{\mathrm{bulge}} = 0.7$), and without galaxy-by-galaxy parameter fitting, this comparison gives an RMS scatter of 0.141 dex and a mean residual of −0.012 dex (Figure 2).

Disk Geometry and QUMOND Numerical Test. For a finite-thickness Miyamoto–Nagai disk, transverse gradients of the algebraic field generate a geometric curl correction. On a 300×300 grid with $a = 3$ kpc and $b = 300$ pc, Biot–Savart reconstruction gives, in this finite-grid configuration, a maximum radial-acceleration correction of 9.23% relative to the one-dimensional algebraic mapping over $a \le r \le 5a$.

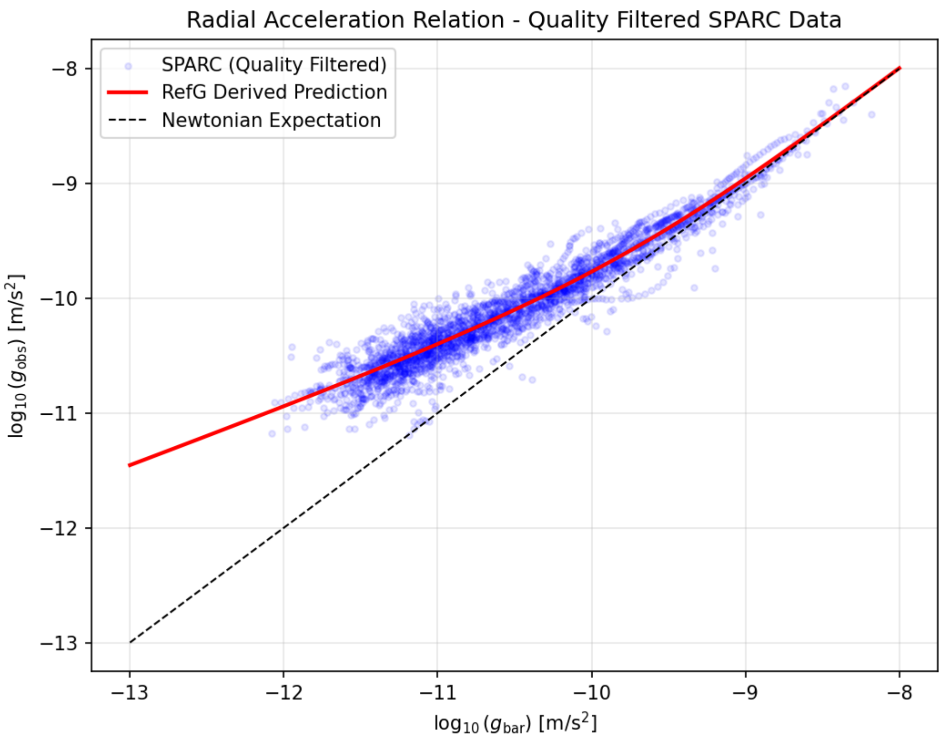

Figure 2: **Radial Acceleration Relation comparison with SPARC data.** The observed radial acceleration $g_{\mathrm{obs}} = V_{\mathrm{obs}}^2/R$ versus the baryonic acceleration g_{bar} for 2,788 selected points across 163 disk galaxies. With fixed a_0 and mass-to-light ratios, but no galaxy-specific fitting, the algebraic RefG curve obtained from $W(g_v, g_N)$ gives an RMS scatter of 0.141 dex and a mean residual of -0.012 dex.

Refractive Gravitational Lensing. Along the flat rotation curve branch, $\Phi_{\text{eff}} = v_f^2 \ln(r/r_0)$ and $n(r) \approx 1 - 2\Phi_{\text{eff}}/c^2$. Integrating the transverse refractive index gradient yields a deflection angle $\hat{\alpha} = 2\pi v_f^2/c^2$, matching the prediction of General Relativity for a singular isothermal sphere. Rotation and lensing thus emerge as two readouts of a single effective potential.

The closest modern analogue to this framework is the **Superfluid Dark Matter** model of Berezhiani and Khoury, where a MOND-like force arises from phonon excitations in a condensate [Berezhiani and Khoury, 2015]. However, while superfluid dark matter introduces an additional particle component, in RefG the coherent flow is a global regime of the unified foundation-medium itself.

Morphology. In RefG, galactic morphology reflects the flow regime of the foundation: ordered laminar rotation is associated with disk and spiral structure, whereas more chaotic or isotropic flow is associated with dispersion-supported elliptical galaxies.

Galaxy Clusters. In galaxy clusters, modifying the acceleration law alone does not fully resolve the dynamical mass deficit (historically highlighted by Fritz Zwicky in the Coma cluster; Zwicky 1933). RefG interprets the cluster mass budget through three distinct channels:

1. The global nodal deficit of the long-wavelength resonant network;
2. Individual galaxies and their local oscillon wakes;
3. Cluster-scale vortical circulation and the hydrodynamic memory of past mergers.

All three enter a unified pressure-deficit budget, each counted exactly once.

Two decisive observational signatures distinguish this framework:

1. **Quantized Circulation:** If the substrate flow possesses a single-valued condensate phase, circulation is quantized in units of:

$$\Gamma = \oint \mathbf{v} \cdot d\mathbf{l} = n\frac{h}{m}, \tag{15}$$

   where m is the effective mass of the flow mode. For ultralight modes ($m \sim 10^{-22}$ eV), vortex filaments could be detected via weak lensing or stellar disk kinematics [Hu et al., 2000, Hui et al., 2021].

2. **The Frozen Memory of the Bullet Cluster:** In the Bullet Cluster (1E 0657-558), weak lensing centroids are displaced from the X-ray gas at 8σ significance, tracking the collisionless galaxies [Clowe et al., 2006]. RefG interprets this separation as the **frozen memory of substrate vortical flow**: during collision, the gas was decelerated by hydrodynamic ram pressure, whereas the substrate deficit trace propagated forward with the galaxies. The memory timescale, spatial morphology, and lensing amplitude form a unified dynamical picture.

Thus, the unified observational program spans SPARC/RAR data [Lelli et al., 2016], cluster mass budgets, Bullet Cluster mergers, the CMB, BAO, and large-scale structure statistics.

3.3 The Cosmological Scale: Foundation Relaxation and Two Languages of Time

At cosmological scales, the external metric description preserves the principal features of observable kinematics: the Universe expands, intergalactic distances grow, and light undergoes cosmological redshift. RefG reads this history from within as the **historical, global relaxation of the foundation-medium**. The cumulative traces of billions of oscillons coalesce into a common background and gradually alter the substrate's mean pressure–energy state. In the process description, early epochs possessed a larger internal-cycle budget for structure formation.

The Coordinate Bridge between Time Measures and Continuity. From the local pressure–phase postulate, $\Phi = -\frac{1}{2}\ln(P/P_{\rm ref})$, the foundation's process time τ and the observer's metric coordinate time t are related by the local cadence factor:

$$\frac{d\tau}{dt} = \Omega = e^{-\Phi} = \sqrt{\frac{P}{P_{\rm ref}}}. \tag{16}$$

The selected cosmological branch is defined by the background pressure law $P(a) = P_{\rm ref}a^{-3/2}$. On this branch, the coordinate transformation takes the exact rational form

$$\frac{d\tau}{dt} = a^{-3/4}, \qquad N = \frac{dt}{d\tau} = a^{3/4}. \tag{17}$$

This bridge preserves cosmological kinematics coherently in both descriptions of time:

1. **Photon redshift:** Comparing wave phases at emission and observation exactly recovers the standard metric law,
$$1 + z = \frac{a_{\rm obs}}{a_{\rm em}}. \tag{18}$$
2. **Agreement of distance measures:** The comoving-distance integrand traversed by light is identical in metric and process time,
$$d_C = \int \frac{c\,dt}{a(t)} = \int \frac{c\,a^{3/4}d\tau}{a(\tau)}. \tag{19}$$
3. **The continuity equation:** The evolution of matter density in metric time, $\dot\rho + 3H_t\rho = 0$, is exactly equivalent to matter conservation in process time, $\rho' + 3H_\tau\rho = 0$.

The Nonlinear Relaxation Family and Metric Acceleration. The selected RefG cosmological family parameterizes the nonlinearity of foundation relaxation by the exponent α. When the process is read in metric t-time through oscillon clocks, the time-stretch factor $N = a^{3/4}$ gives the Hubble rate a nonlinear dynamical form:

$$H_t(a) = A\,a^{-3\alpha/2} + B\,a^{-3/2}, \tag{20}$$

where $A = \frac{2}{3}\kappa_0$ is the vacuum-relaxation amplitude, B is the matter contribution, and α is the nonlinear-relaxation exponent. In process time τ, the rate of the vacuum term is proportional to $(P/P_{\rm ref})^{\alpha+1/2}$; hence $\alpha = -1/2$ is the special linear branch on which vacuum relaxation proceeds at a constant rate. The free-α branch generalizes this special case.

Two Measures of Cosmic Age and the Process Budget of the Early Universe. Integrating with the observer's metric clock gives

$$t_0 = \int_0^1 \frac{da}{aH_t(a)} \approx 13.62 \text{ Gyr}. \tag{21}$$

This value lies numerically close to astrophysical age bounds from globular clusters and to the 13.8 Gyr age inferred under Planck-calibrated ΛCDM [Planck Collaboration, 2020].

The process interval of the same cosmological history is

$$\tau_0 = \int_0^1 \frac{a^{-3/4}\,da}{aH_t(a)} \approx 29.86 \text{ Gyr}. \tag{22}$$

This is a present-clock-calibrated measure of the evolutionary cadence accumulated within the same history. The higher background pressure of the early Universe accelerates process cadence, $d\tau/dt \propto (1+z)^{3/4}$, so that far more internal cycles fit within a given metric interval. The early massive and luminous galaxies observed by the James Webb Space Telescope [Labbé et al., 2023] provide a direct and stringent test of this mechanism: the equations governing density-perturbation growth, gravitational collapse, and star formation must show how the additional process budget translates into observable structural growth.

Observational Test and Statistical Comparison. A joint fit was performed to the Pantheon Type Ia supernova sample (40 bins with the full systematic covariance; Scolnic et al. 2018) and to cosmic chronometers (CC: 31 measurements of $H(z)$):

- **Nonlinear** RefG **model:** $\chi^2_{\text{tot}} = 53.75$ (CC: 14.44, SN: 39.32), with best-fit parameters $H_0 = 69.07\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, $\Omega_A = 0.452$, and $\alpha = -0.673$.
- **Standard ΛCDM model:** $\chi^2_{\text{tot}} = 53.94$ (CC: 14.62, SN: 39.32), with $H_0 = 69.07\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ and $\Omega_m = 0.299$.

In raw goodness of fit, RefG is marginally ahead of ΛCDM ($\Delta\chi^2 = -0.19$). Accounting for the additional parameter α through Occam's razor and the information criteria gives

$$\Delta AIC = +1.81, \qquad \Delta BIC = +4.07. \tag{23}$$

On these data, AIC and BIC give ΛCDM a modest advantage. Figure 3 displays the two observational comparisons entering the joint fit. The physical distinction of RefG lies in its mechanism: accelerated expansion is described by dynamical relaxation of the foundation rather than by a constant Λ.

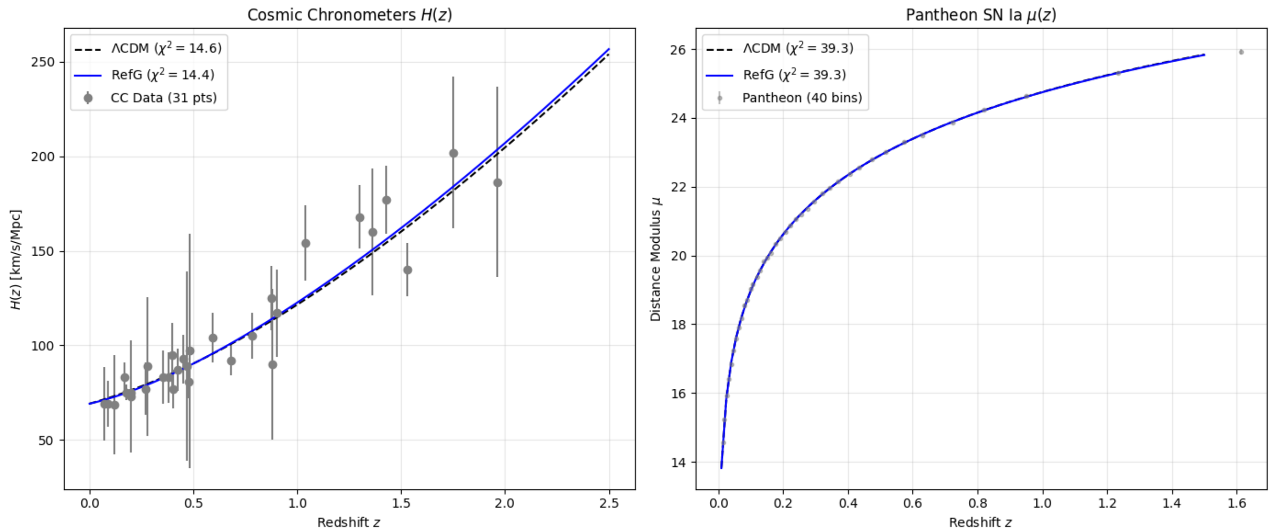

Figure 3: **Joint cosmological fit of the nonlinear** RefG **branch and ΛCDM.** The left panel shows $H(z)$ from 31 cosmic-chronometer measurements; the right panel shows the distance modulus of the 40-bin Pantheon Type Ia supernova sample. The solid blue curve is the free-α RefG branch, and the dashed black curve is ΛCDM.

Future Kinematics: Phantom Acceleration and the Big Rip. The best-fit exponent, $\alpha \approx -0.673$, gives the late-time asymptotic limit

$$q_\infty = -1 + \frac{3}{2}\alpha \approx -2.009. \tag{24}$$

Because $q < -1$, this is the kinematic analogue of phantom dark energy. If the free-α law is extrapolated unchanged, expansion reaches a Big Rip in a finite metric future:

- remaining metric time: $t_{\text{rip}} \approx 24.46$ Gyr;
- remaining internal evolutionary time: $\tau_{\text{rip}} \approx 12.59$ Gyr.

On the internal-cycle measure of this unsaturated branch, approximately 70.3% of the total interval to the Big Rip has already elapsed. A transition to a stable, de Sitter-like branch requires a late-time relaxation law for which $\dot{P}/P \to -\text{const}$ and hence $q \to -1$. Determining that late-time law is the next stage of the cosmological model.

Ontological Consistency and the Arrow of Time. Within the ontology of a unified medium, the familiar **vacuum-energy problem**—in which the sum of quantum zero-point fluctuations exceeds the observed scale associated with Λ by some 120 orders of magnitude—takes a different accounting form. An infinite sum over the energies of independent fields is no longer the foundational ledger, because all channels participate in the common equilibrium of one self-sustaining medium. The small cosmic acceleration is the residual dynamical departure from that equilibrium. Related mechanisms have been developed in the q-theory program of Volovik and Klinkhamer [Volovik, 2003, Klinkhamer and Volovik, 2008].

RefG identifies the macroscopic **arrow of time** with the global direction of foundation relaxation. Like a colored crystal dissolving in water, energy dispersed through the medium does not spontaneously reassemble into a solid, localized crystal. This statistical irreversibility [Planck, 1899] unifies the thermodynamic, cosmological, and radiative arrows of time.

3.4 Quantum Randomness and Two-Level Time

Two-level time provides a clarifying lens for quantum phenomena. The spacetime coordinates of a measurement event are recorded in oscillon (clock) time, while the selection of a specific outcome is associated with the deeper, pre-clock state of the foundation. This pre-clock level cannot be measured directly by clocks, because clocks are themselves emergent oscillon systems.

Bell's theorem and its experimental verifications rule out local hidden-variable theories [Bell, 1964, Hensen et al., 2015]. Meanwhile, the **no-signaling theorem** ensures that entanglement correlations cannot transmit superluminal messages.

RefG interprets this as **atemporal internal coordination**: within the internal language of the foundation, entanglement is not a physical signal propagating through space; in the metric language of clocks, it appears as an instantaneous statistical correlation between separated measurement outcomes.

Wheeler's **delayed-choice experiment** is understood in the same terms [Wheeler, 1978, Jacques et al., 2007]: a delayed measurement choice determines whether wave interference or which-way particle information is realized. RefG's two-level time describes this without retrocausal signaling, as an expression of the underlying phase integrity of the state.

Related concepts in modern physics include:

- The atemporal ground level of the **Wheeler–DeWitt equation** ($\hat{H}\Psi = 0$; DeWitt 1967);
- The **Page–Wootters mechanism**, where time emerges from subsystem quantum correlations [Page and Wootters, 1983];
- **de Broglie–Bohm pilot-wave theory**, characterized by non-local phase coordination [Bohm, 1952].

This quantum framework is governed by two strict quantitative requirements:

1. **Born's Probability Rule ($P = |\psi|^2$):** The model must derive the observed statistical distribution of measurement outcomes [Born, 1926];
2. **The No-Signaling Condition:** Internal phase coordination must never generate a transmissible signal that violates relativistic causality.

4 Oscillons and the Origin of Matter

4.1 Many Patterns in One Ocean

To build physical intuition, imagine a single continuous ocean and the diverse wave motions that arise within it. Most waves form, propagate across the surface, and quickly dissipate; under specific nonlinear conditions, however, certain configurations self-organize into stable, localized, self-sustaining structures:

- One may form a toroidal vortex ring, with fluid circulating continuously in a closed loop;
- Another may manifest as a stable spherical pulsation (soliton);
- A third may take shape as a helical, twisting wavepacket.

This hydrodynamic picture illustrates the foundational thesis of this chapter: **Electrons, protons, photons, and quarks are distinct resonant and topological patterns supported by a single universal foundation-medium.** The continuous medium itself oscillates: massive particles are locally confined modes of this oscillation, whereas photons are freely propagating transverse wave states.

A historical precursor is Tony Skyrme's model, where nucleons (protons and neutrons) are described as topological solitons—"skyrmions"—of a nonlinear meson field [Skyrme, 1961]. RefG elevates this intuition to the fundamental ontological level: all such soliton-like structures are normal modes of a single universal foundation-medium.

The classical image of a rigid, pointlike particle gives way to a **continuous wave-medium ontology**: spatial extent is characterized by the vibrational amplitude and phase of the foundation, while a localized event registered in a detector represents the resonant coupling between this extended vibration and the internal oscillon modes of the detector's atoms. RefG thus grounds the spatial amplitude and phase of the quantum wave function in an explicit ontological substrate.

In an atom, the "electron cloud" is an actual vibrational amplitude distribution rather than the rapid orbital transit of a point mass. The extended state maintains a coherent phase structure throughout space, and the localized point of measurement emerges through local resonant locking with the detector.

The single-electron double-slit experiment synthesizes two foundational realities: the wave amplitude depends simultaneously on the global geometry of both slits, yet each detection event on the screen manifests as a discrete, localized point. Akira Tonomura's group directly recorded this progressive statistical accumulation using electron microscopy [Tonomura et al., 1989]. RefG interprets an atom's stationary electronic state as a standing, confined wave mode of the medium rather than an accelerating classical point charge; consequently, electromagnetic radiation occurs only during transitions between discrete states.

The spatial spread of a wavepacket and the intrinsic structural form factor of a particle are distinct physical concepts:

- Elastic scattering probes charge and energy distribution form factors, while deep inelastic scattering reveals internal parton structure and distribution functions;
- The spatial extent of a propagating wavepacket does not enter these observables as an intrinsic physical size.

Empirically, the electron exhibits a pointlike electromagnetic form factor down to scales of order 10^{-18} m [Navas et al., 2024]. A direct empirical test of the oscillon framework is whether the form factor computed from its internal topological charge profile reproduces this pointlike behavior across accessible momentum transfer scales.

4.2 Localized Confinement: Massive and Massless Modes

In the RefG particle framework, the defining ontological dividing line between massive and massless sectors is **localized confinement**.

Massive Oscillons. The energy of a massive particle is confined within a localized, self-sustaining configuration. When a wavepacket is trapped via vortex circulation or stable nonlinear pulsation within a finite spatial region, Bernoulli-type dynamics induce a persistent static pressure deficit in the surrounding medium.

Long-lived localized nonlinear pulsations are well-known in classical field theory as **pulsons and oscillons** [Bogolyubsky and Makhankov, 1976, Gleiser, 1994, Copeland et al., 1995]. RefG proposes that their stability arises from population-level resonant locking and conserved topological charges. In external readout, this locally confined energy manifests as inertia and effective gravitational mass.

While standard field-theoretic oscillons eventually decay through radiation leakage, RefG proposes nonlinear phase-locking with the surrounding particle population as the mechanism that suppresses radiative loss and selects long-lived particle states.

Massless Carriers (The Photon). A massless mode possesses neither a rest frame nor a localized static core—it propagates through the network at the limiting phase transmission velocity (c). RefG interprets the photon as a freely propagating transverse phase mode.

Helicity and localized rotation are fundamentally distinct phenomena:

- The ± 1 helicity of a photon describes the spatial spirality (handedness) of transverse electric and magnetic phase vectors along the propagation axis, without forming a localized static vortex;
- A massive mode possesses a localized topological core and a static pressure-deficit well.

The energy-momentum tensor $T_{\mu\nu}$ of a photon couples fully to the gravitational energy balance: while free electromagnetic radiation does not create a static, localized deficit well like a massive oscillon, its propagating energy enters the total gravitational source of the system.

The same principle applies to composite hadrons: inside protons and neutrons, the confined energy of strong quark and gluon interactions does not radiate an external wake, but contributes fully to the hadron's total inertial energy and externally read gravitational mass.

Local topological form, spin, electric charge, color, and polarization preserve internal quantum distinctions, while gravitational readout unifies them through a single pressure-deficit channel.

4.3 Self-Forming Resonators and the Two-Level Filter

Nonlinear wave equations admit an infinite diversity of mathematical solutions; nature, however, realizes a sparse, strictly defined spectrum of stable elementary particles. Why does nature select this specific discrete set?

RefG organizes this proposed selection mechanism through a **two-level dynamical filter**:

1. Level One: Kinematic and Population Filter. An acoustic handpan provides an intuitive analogy. While the resonant length of a guitar string is externally fixed by its wooden body, a three-dimensional oscillon must sculpt its own resonant cavity within the medium. On a handpan's surface, a stable acoustic note unites the fundamental frequency, its octave, and the compound fifth into a harmonic $1:2:3$ triad.

This triad is a structural rhythm: a self-forming resonator binds several mutually coherent modes into a single physical entity. In RefG, an oscillon's pressure deficit alters the local geometry, which in turn filters viable modes: a stable mode must propagate through the substrate while locking into resonant coexistence with the surrounding particle population.

2. Level Two: Energetic Stability Filter. Any mode that passes the kinematic filter but possesses an energetically accessible decay channel rapidly disintegrates into lighter, stable ground states.

Hadronic resonances (Δ, ρ, ω) illustrate this: with typical lifetimes of 10^{-22}–10^{-24} s, they appear in scattering spectra but cannot endure as asymptotic particles, shedding energy into stable channels (pions, protons).

Electron decay has never been observed; as the lightest charged lepton, it is protected by electric charge conservation. The extraordinary experimental stability of the proton ($\tau_p > 10^{34}$ years) is linked in RefG to corresponding topological protection.

The mathematical language governing these filters is that of nonlinear **attractors**—stable phase-space configurations toward which dynamical trajectories asymptotically converge.

A stable particle represents a self-consistent equilibrium: the wave configuration carves a pressure deficit in the medium, and this deficit acts as a potential well that stabilizes and reinforces the configuration. In classical field theory, Sidney Coleman's Q-balls are protected by conserved internal Noether charges [Coleman, 1985]. In RefG, stability is reinforced by nonlinear phase-locking across the entire population.

This collective synchronization requires no external coordinator: Christiaan Huygens's classic observation of pendulum clocks synchronizing through a common wooden beam [Huygens, 1665], mode-locking in lasers, and Yoshiki Kuramoto's nonlinear synchronization model [Kuramoto, 1975] demonstrate how collective order emerges endogenously. In particle physics, this vision resonates with Geoffrey Chew's S-matrix "Bootstrap" program, where particle spectra form a self-consistent whole [Chew and Frautschi, 1961].

Under adiabatic gravitational changes, resonant locks are preserved: the collective cadence $\Omega(x)$, operational scale, and externally read mass adjust synchronously, while dimensionless ratios and phase harmonies remain invariant. Rapid, high-energy shocks can break this locking, redistributing energy into other compatible channels.

4.4 The Global Resonator: Unity of Micro- and Macro-Spectra

The two-level filter specifies the proposed selection mechanism; we now ask where it is realized.

Ernst Chladni's classic acoustic experiment provides intuition [Chladni, 1787]: sand scattered across a vibrating metal plate gathers along nodal lines where vibrational amplitude vanishes. In RefG, global modes and their nodal boundaries govern this distribution: every resonator possesses an intrinsic spectrum, with its geometry and boundary conditions determining which modes are amplified and which are damped.

RefG conceptualizes the Universe as a **unified global volumetric resonator**—a three-dimensional foundation-medium endowed with an intrinsic allowed spectrum. The primordial excitation of the cosmological origin populated this spectrum across all scales:

- **Short-Wavelength Modes** (high frequencies) manifest as localized elementary particles (oscillons);
- **Long-Wavelength Modes** (low frequencies) govern the large-scale distribution of matter: the **Cosmic Web**.

Galaxies and galaxy clusters resemble sand on a Chladni plate: their spatial architecture traces the nodal surfaces, filaments, and vast cosmic voids of the foundation's global standing-wave modes.

Standard cosmology describes this large-scale structure via Yakov Zel'dovich's anisotropic gravitational collapse [Zel'dovich, 1970]. RefG maps this morphology onto the spectral properties of a single operator.

The central hypothesis of this map states: **Microscopic elementary particles and the macroscopic cosmic web are two distinct readouts of a single universal spectral operator**—the former as locally bound topological solitons, the latter as globally extended cosmological wave patterns.

A unified empirical test of this operator would have to encompass:

1. The discrete mass spectrum of elementary particles;
2. The cosmological history of structure growth;
3. The mass budgets and dynamics of galaxy clusters;
4. The Baryon Acoustic Oscillation (BAO) standard ruler at $\sim 100\,h^{-1}$ Mpc [Eisenstein et al., 2005].

Within a finite, boundaryless resonator (conceptually related to Christof Wetterich's variable-gravity framework; Wetterich 2013), cosmological expansion describes the evolving relationship between substrate geometry and its internal material standards, eliminating the need for an external embedding container.

In contemporary Planck data, low-multipole CMB anomalies ($l = 2, 3$) remain moderate, and cosmic topology searches set lower bounds on spatial volume without singling out a specific global topology [Planck Collaboration, 2016].

4.5 Discreteness and the C_3 Generation Symmetry

In the Standard Model, the masses of the charged leptons—electron (e), muon (μ), and tau (τ)—are arbitrary Yukawa coupling inputs. RefG treats them as the **ninth-order internal three-phase (C_3) triad** of a single topological defect: three distinct resonant modes of one underlying physical structure.

The charged-lepton mass parametrization utilizes the vibrational scaling $m \propto \nu^2$ (or $\nu \propto \sqrt{m}$) within a cyclic three-phase geometry:

$$\nu_k = \mu \left[1 + A_K \cos\left(\theta + \frac{2\pi k}{3}\right)\right], \qquad k = 0, 1, 2, \tag{25}$$

where:

- μ is an overall frequency scale;
- A_K is the dimensionless three-phase modulation amplitude;
- θ is the mass phase angle;
- $k = 0, 1, 2$ indexes the three generations (e, μ, τ).

When $A_K = \sqrt{2}$, the C_3 algebra identically recovers Yoshio Koide's empirical mass relation [Koide, 1982, 1983]:

$$K = \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{2}{3}. \tag{26}$$

This three-phase parametrization possesses a rich literature:

- Carl Brannen formulated the relation in terms of circulant matrix eigenspaces [Brannen, 2006];
- Robert Foot demonstrated that $K = 2/3$ corresponds to a 45° geometric orientation in $\sqrt{m}$-space [Foot, 1994];
- Brannen's Pancharatnam–Berry phase construction yielded $\pi/12$ while leaving $\theta = 2/9$ as a dynamical parameter [Brannen, 2010];
- Konstantin Shulga obtained $\theta_\ell = -2/9$ within a compact family cycle [Shulga, 2026].

RefG offers a candidate route connecting this structure to a **ninth-order holonomic return cycle**:

1. On the selected $h = 2$ branch, the first non-trivial topological return gives the candidate mass angle:
$$\theta = \frac{h}{9} = \frac{2}{9} \text{ rad.} \tag{27}$$
2. $A_K = \sqrt{2}$ represents energetic equipartition between internal singlet and doublet vibrational channels.

A simple C_3-invariant potential exhibits **θ-blindness**—an inability to select θ dynamically—showing that a simple potential minimum is insufficient and motivating closed topological holonomy as a candidate origin of the mass angle.

Evaluating $(A_K, \theta) = (\sqrt{2}, 2/9)$ at tree level, anchored to the physical electron pole mass, yields deviations of 451.7σ for the muon mass and 0.61σ for the tau mass. Reconciling this tree-level geometry with measured pole masses requires an action-derived radiative correction bridge [Sumino, 2009].

This structure has a clear acoustic analogue: **The C_3/Koide triad of charged leptons is an anharmonic, stable chord supported by a 3D volumetric resonator**; within the candidate map, all three generations are normal modes of a single topological defect.

5 Topology and Quantum Numbers

5.1 Phase, Orientation, and Topological Quantization

Thus far, the foundation-medium has been described primarily in terms of vibrational amplitude and local pressure. However, the complete local state of the medium is characterized by multiple coupled field variables: the phase-clock scalar, amplitude profile, and spatial orientation frame each possess distinct degrees of freedom. In this monograph, Φ represents the compact designation for the complete state of the foundation-medium rather than a simple single-component scalar: its full configuration simultaneously carries amplitude, phase, internal geometric form, and spatial orientation.

If Φ were merely a single-valued scalar field, its velocity field would be defined strictly by the gradient $\mathbf{u} = \nabla\Phi$, forcing the curl (vorticity) to vanish identically ($\nabla \times \mathbf{u} = 0$). The emergence of a non-trivial topological sector requires:

- **A multi-valued phase**,
- **A singular defect nodal line**, and
- **A localized internal spatial orientation frame**.

At the central core of a defect line, the wave amplitude vanishes ($|\psi| = 0$), while around it the phase winds by an integer multiple of 2π ($\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n$). A canonical laboratory analogue is a quantized vortex filament in superfluid Helium.

The cosmological analogue is a **cosmic string**—a macroscopic topological defect line formed during a symmetry-breaking phase transition in the early Universe via the Kibble mechanism [Kibble, 1976]. In RefG, this represents a "frozen seam" of the foundation: vacuum topology determines whether such lines form, while astrophysical observations (CMB lensing and pulsar timing arrays) tightly constrain their cosmological tension.

Lord Kelvin's seminal papers on vortex atoms and vortex motion [Thomson, 1867, 1869] inspired Peter Guthrie Tait to systematically catalog knots, laying the foundations of modern mathematical knot theory [Tait, 1877]. RefG continues this lineage using the modern vocabulary of pressure-energy balance and population-locked resonance.

History provides a crucial lesson: 19th-century vortex atoms in ideal inviscid fluids failed to yield a stable discrete spectrum or reproduce atomic spectra. In RefG, the topological knot provides the kinematic scaffolding of form, while physical viability and dynamical stability are secured by pressure-energy feedback and nonlinear phase-locking across the oscillon population.

Phase quantization occurs when the field is single-valued and continuous along a closed loop, returning to its initial state modulo 2π, with the loop belonging to a non-trivial first homotopy group ($\pi_1 \neq 0$).

For a closed, orientable, framed ribbon, the **Călugăreanu–White–Fuller (CWF) theorem** applies [Călugăreanu, 1961, White, 1969, Fuller, 1971, Ricca and Nipoti, 2011]:

$$Lk = Wr + Tw, \tag{28}$$

where:

- Lk (Linking Number) is the topological integer invariant measuring total linking;
- Wr (Writhe) measures the 3D spatial coiling of the ribbon's central axis;
- Tw (Twist) measures the internal phase winding of the ribbon frame around its axis.

Under continuous deformation, Wr and Tw can vary continuously (e.g., spatial coiling converting into internal twist), yet their algebraic sum Lk remains strictly invariant.

In fluid dynamics, kinetic helicity ($H = \int \mathbf{u} \cdot (\nabla \times \mathbf{u})\, d^3x$) is an invariant of ideal barotropic flow, which Keith Moffatt connected to the topological linking number Lk [Moffatt, 1969]. In superfluid Helium, circulation quantization was predicted by Lars Onsager and Richard Feynman [Onsager, 1949, Feynman, 1955] and experimentally verified by Joe Vinen [Vinen, 1961]. Integer phase winding is thus an established laboratory reality.

The core principle **"Topological Confinement = Conserved Charge"** implies in RefG that electric charge is the external readout of a closed phase-orientation winding: topological invariants define the discrete charge class, while field dynamics govern its electromagnetic coupling.

In the RefG synthesis, **spin-1/2 and elementary electric charge** are read as two candidate geometric traces of a common double-covered topological source: the half-twist determines spinorial transformation under rotation, while chiral orientation and integer winding label the charge class and sign. Their quantitative completion requires separate derivations of the spin transformation law and electric-flux normalization.

Direct experimental analogues exist: Kelvin waves on quantized vortices in superfluid Helium have been directly excited and their dispersion measured [Minowa et al., 2025]; in classical water vortices, helical axial modes and their dispersion relations have been mapped experimentally [Barckicke et al., 2026]. In RefG, an oscillon's radial breathing mode corresponds to its Compton mass pulsation, while its helicoidal mode governs charge and radiative channels.

5.2 Spin and Fermionic Statistics

A phase field twisted by a half-turn (π) along an oscillon's internal axis generates a **Möbius-strip-type double-covered Z_2 topology**:

- A single spatial rotation of 2π reverses frame orientation, flipping the sign of the wave amplitude ($\psi \to -\psi$);
- A double rotation of 4π fully restores the initial quantum state ($\psi \to +\psi$).

This Z_2 geometry embodies the fundamental transformation property of the **spinor representation ($SU(2)$)**. RefG associates electron spin-1/2 with this internal double-covering. Changing this topological class requires tearing the configuration: vanishing amplitude at the core, defect reconnection, or altering boundary conditions.

Completing the connection to Fermi–Dirac statistics requires that exchanging two identical oscillons produces a phase shift of π, yielding an antisymmetric multi-particle state in Fock space:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1). \tag{29}$$

Under the required exchange antisymmetry, the amplitude for two identical fermions occupying the same state vanishes identically ($\Psi = -\Psi \implies \Psi = 0$). Deriving this exchange phase from the Möbius topology is the remaining step needed to recover the **Pauli Exclusion Principle**. For bosonic modes, the Möbius half-twist is absent, the state remains symmetric under exchange ($\Psi \to +\Psi$), and macroscopic Bose condensation is permitted.

In relativistic quantum field theory, the Spin-Statistics Theorem unites Lorentz invariance, positive energy, and microcausality [Pauli, 1940]. RefG's topological framework offers a candidate geometric mechanism for this connection.

5.3 Charge, the Photon, and the Electromagnetic Channel

RefG defines **electric charge** as the external spatial trace of an oscillon's internal chiral phase twist:

- Right-handed chiral twist corresponds to positive charge ($+e$);
- Left-handed chiral twist corresponds to negative charge ($-e$).

Externally, this trace extends radially in all directions. In quantitative electrodynamics, its surface integral must satisfy Gauss's Law:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0}. \tag{30}$$

A **positron** is the exact topological mirror image of an electron (possessing opposite chiral twist). In electron-positron annihilation ($e^+ + e^- \to 2\gamma$), opposite topological windings untwist and cancel, releasing locally confined energy into freely propagating transverse radiation (photons). In a neutral oscillon, opposing chiral twists cancel internally, yielding zero net flux (characterizing the neutrino sector).

A uniform rotation of the internal phase angle corresponds to global $U(1)$ gauge symmetry. Describing spacetime-dependent rotations necessitates a local frame connection and covariant derivative. RefG identifies the **photon** as a **transverse, helicoidal wave** of this connection: it propagates phase-twist disturbances across space, acting as the massless gauge boson of electromagnetism.

The topological nature of electromagnetism is manifest in classical electrodynamics: Antonio Rañada's vacuum solutions exhibit linked, knotted electric and magnetic field lines classified by the Hopf invariant [Rañada, 1989].

The sign of physical forces (attraction vs. repulsion) is governed by gauge boson spin, source type, and coupling structure:

- **Vector Gauge Channel (Spin-1):** Like charges repel, opposite charges attract;
- **Tensor Gravitational Channel (Spin-2):** All positive energy densities attract universally.

In three spatial dimensions, a conserved massless flux dilutes as $1/r^2$, yielding a $1/r$ potential.

Hydrodynamic Bjerknes forces [Bjerknes, 1906] illustrate the limits of simple fluid analogies: in an acoustic medium, bodies pulsating in-phase attract, while out-of-phase bodies repel—the exact opposite of electrostatics. This demonstrates that force signs are not reducible to crude mechanical collisions; they are governed by the rigorous mathematics of vector gauge fields, source currents, and topological linking.

The fine-structure constant $\alpha \approx 1/137$ sets the dimensionless coupling strength of this electromagnetic channel and is interpreted as the impedance ratio of the substrate's electric-flux and transverse-photon response channels.

5.4 Color Confinement and Three Generations

The three-dimensionality of space ($D = 3$) plays a crucial structural role in the RefG particle map: **The three-axis spatial orientation frame of an oscillon provides the geometric substrate for the color charge triad (red, green, blue: r, g, b).** Color represents the internal spatial degree of freedom of this frame:

- **Quark:** An oscillon mode vibrating along a single internal axis is topologically open, possessing unconfined endpoints (an open topological segment);
- **Baryon (Proton, Neutron):** The symmetric combination of all three axes ($r + g + b$) forms a fully closed, stable 3D topological knot (a color-singlet white state);
- **Meson:** Pairing an axis with its opposite anti-axis (color + anticolor) likewise forms a closed, color-neutral loop.

RefG identifies the candidate mechanism for **color confinement in Quantum Chromodynamics (QCD)** with the geometric incompleteness of an isolated quark: a closed, gauge-invariant state cannot be formed from a single open axis.

The mode count of an unconstrained three-axis deformable oscillon matches the 8 generators of $SU(3)$, establishing the structural map of the color sector. Formal identity requires verifying the $SU(3)$ structure constants, local gauge invariance, and interaction vertices.

The **Three-Generation Puzzle** is a proposed extension of the C_3 structure explored in Chapter IV:

- Charged leptons (e, μ, τ) represent the ninth-order three-phase vibrational modes of a single topological defect;
- Quark generations $(u/d, c/s, t/b)$ replicate this generational architecture across the three-axis colored sector.

In this candidate map, the three generations are read as successive holonomic return modes of a single 3D volumetric resonator.

The universal foundation of holonomy is **Berry's geometric phase** [Berry, 1984]: under adiabatic cyclic transport across parameter space, a quantum state acquires an invariant geometric holonomy phase alongside its dynamical phase. RefG links this to the ninth-order return cycle of a framed defect, yielding $\theta = 2/9$ rad on the $h = 2$ branch.

Why is Space Three-Dimensional? In RefG, $D = 3$ is intimately tied to the **stability of topological knots**:

- Closed one-dimensional loops form stable, non-trivial knots and links exclusively in $D = 3$ spatial dimensions;
- In higher dimensions $(D \geq 4)$, any knotted loop can be untied and unlinked smoothly along extra spatial dimensions without self-intersection.

This topological argument builds upon the insights of Paul Ehrenfest, Gerald Whitrow, and modern investigators [Ehrenfest, 1918, Whitrow, 1955, Berera et al., 2017]. RefG formulates the co-emergence of material discreteness and 3D space from a single topological locking mechanism as a central working hypothesis.

5.5 Geometric Map of the Standard Model

RefG organizes the particle spectrum of the Standard Model into the following geometric architecture:

1. The Electron and Charged Leptons. The electron is a complete toroidal, framed Möbius Z_2 topological defect with all three spatial axes locally closed. RefG connects this double-covered geometry to the Dirac factor $g = 2$.

The electron's measured magnetic moment slightly exceeds 2. Julian Schwinger calculated the leading quantum electrodynamic correction $(\alpha/2\pi)$, while modern Penning trap experiments measure $|g_e|/2 = 1.00115965218059(13)$ [Schwinger, 1948, Fan et al., 2023]. This precision measurement provides a benchmark for testing the quantum completion of the topological map.

2. Quarks and Color Charge. Quarks are open axial topological segments. Their fractional electric charges $(+2/3, -1/3)$ are associated with one-third subdivisions of the 3D closed loop; their exact values remain to be fixed by canonical charge normalization. Color confinement dictates that only closed, color-singlet configurations (baryons and mesons) exist as asymptotic states.

3. Gauge Bosons.

- **Photon:** Massless propagating transverse helicoidal wave ($U(1)$ gauge symmetry, 2 helicities);
- **Gluons:** 8 massless transverse modes mediating color transitions between internal axes ($SU(3)$ octet);
- $W^{\pm}$ **and** Z^0 **Bosons:** Locally confined, massive carrier oscillons mediating electroweak transitions ($SU(2)_L \times U(1)_Y$).

RefG interprets **beta decay** ($n \to p + e^- + \bar{\nu}_e$) as a topological reconnection: altering the internal axis orientation ($d \to u$) produces a transient charged weak carrier (W^-) that decays into leptonic channels, bringing chirality, energy–momentum balance, and the decay spectrum into a single geometric transition.

4. Neutrinos and the Higgs Boson.

- **Neutrinos:** Electrically neutral, incompletely confined chiral phase modes. Their incomplete topological confinement is associated with their ultra-small masses ($m_\nu < 0.1$ eV) and weak coupling, while Pontecorvo flavor oscillations represent phase interference among mass eigenspaces;
- **Higgs Boson:** A purely spherical, radial breathing pulsation of the foundation-medium devoid of angular momentum, naturally corresponding to a spin-0 scalar state ($m_H \approx 125.25$ GeV).

5. The Graviton and the Weinberg–Witten Theorem. The tensor wave sector contains two transverse-traceless polarizations ($h_+, h_\times$). The Weinberg–Witten theorem [Weinberg and Witten, 1980] forbids massless spin-2 particles with a Lorentz-covariant, gauge-invariant energy-momentum tensor in flat spacetime. At the low-energy effective level, RefG can meet this constraint only if the same branch simultaneously carries a massless tensor mode, universal coupling, and effective Lorentz symmetry. The graviton is then read not as an isolated fundamental particle on a fixed background, but as a collective tensor-shear excitation of the foundation-medium.

6. Nonlinear Soliton Lagrangian. The formal mathematical foundation of this topological map relies on nonlinear field Lagrangians admitting stable knotted soliton solutions (in the spirit of the Faddeev–Niemi model; Faddeev and Niemi 1997).

5.6 Fundamental Constants: α and the Gravitational Hierarchy

The fine-structure constant:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.035999} \tag{31}$$

remains one of the deepest mysteries in physics, described by Richard Feynman as one of the greatest unexplained numbers in science [Feynman, 1985].

In RefG, α is interpreted as the **dimensionless impedance ratio of the substrate's electric-flux and transverse-photon response channels**.

This definition is governed by three precise distinctions:

1. Simple ratios of Planck and electron masses or frequencies do not yield 137;
2. The identity $r_e/\bar{\lambda}_c = \alpha$ is an algebraic consequence of dimensional definitions;
3. Integer phase-locking steps represent physical realities (as demonstrated by the Josephson effect and Shapiro steps; Josephson 1962, Shapiro 1963), though seconds are themselves calibrated by atomic oscillons.

Thus, α is not an absolute frequency or energy; it is the **dimensionless impedance ratio of the substrate's electric-flux and transverse-photon response channels**. Under local pressure shifts, both responses scale synchronously, leaving α invariant.

The Hierarchy Problem: Why is Gravity So Weak? The electrostatic repulsion between two protons exceeds their gravitational attraction by a factor of 10^{36}, while for electrons this ratio reaches 10^{42}:

$$\frac{F_{\rm em}}{F_g} = \frac{e^2}{4\pi\epsilon_0 G m_e^2} = \alpha \left(\frac{M_{\rm Pl}}{m_e}\right)^2 \sim 4.17 \times 10^{42}. \tag{32}$$

This immense disparity motivated Paul Dirac's Large Numbers Hypothesis [Dirac, 1937].

RefG interprets this hierarchy through the **differing ontological character** of the two interactions:

1. **Electromagnetism** is a primary, direct transverse-phase gauge coupling between topological charges;
2. **Gravity** is not a primary force—it is the **secondary, Bernoulli-type refractive pressure deficit** ($-\Delta P \propto \rho_{\rm dyn}$) induced in the substrate by an oscillon's internal oscillation.

The Planck mass ($M_{\rm Pl} = \sqrt{\hbar c/G} \approx 2.18 \times 10^{-8}$ kg $\approx 1.22 \times 10^{19}$ GeV) defines the saturation stiffness scale of the foundation-medium. Electrons and protons are localized oscillon modes possessing energies many orders of magnitude below this scale ($m_e/M_{\rm Pl} \sim 10^{-22}$).

Because the gravitational pressure deficit is proportional to oscillon energy (m_e), and gravitational attraction couples two such deficits (Gm_e^2), gravitational interaction between elementary particles is quadratically suppressed relative to the Planck scale ($m_e^2/M_{\rm Pl}^2 \sim 10^{-44}$).

Thus, in RefG, the extreme weakness of gravity reflects its secondary, refractive nature. The ratio $\alpha(M_{\rm Pl}/m)^2$ captures this scaling; deriving the precise numerical values of G and particle masses from first principles remains an active goal.

6 Epilogue: Boundaries of Knowledge

6.1 Why One Foundation and One Manifested Medium?

At the conclusion of this work, we return to the central physical and philosophical question: **Why a single foundation-medium?**

One primary motivation is ontological parsimony (Occam's razor): a single substrate radically minimizes the number of independent, disconnected axiomatic inputs. However, mathematical simplicity alone is insufficient to establish physical truth.

A far deeper and more decisive argument is the **common origin of physical law**. When electrons, quarks, and photons are understood as distinct resonant and topological modes of a single universal foundation-medium, RefG seeks to derive the conservation of energy, momentum, angular momentum, and electric charge from the dynamic equilibrium of this common substrate rather than postulating them as external, unconnected axioms. Physical interactions represent couplings between differing modes of the same medium, while conservation laws express the geometric continuity of its total energy balance.

This perspective addresses Eugene Wigner's famous reflection on the "unreasonable effectiveness of mathematics" in the natural sciences [Wigner, 1960]. The postulate of a single ontological substrate provides this profound harmony with a physical explanation: **The universal mathematical order of physical law reflects the internal dynamical coherence of a single primary substance.**

Postulating multiple mutually independent fundamental fields (as in the Standard Model) merely defers the question: *What overarching symmetry principle constrains these disparate fields to obey the same universal conservation laws and relativistic spacetime symmetries?* Within the RefG ontology, this universality is not an external decree—it is the intrinsic dynamics of the shared substrate.

This argument gives the Standard Model's **fine-tuning problem** a common structural target:

- The stringent observational bound on the electrical neutrality of bulk matter ($|q_p + q_e|/e < 10^{-21}$);
- The observational compatibility of the photon rest mass with zero ($m_\gamma < 10^{-18}$ eV);

- The precise quantization of fractional quark charges [Navas et al., 2024].

These interrelated constraints call for a common structural explanation, which RefG associates with the topological geometry of the foundation-medium.

A single universal foundation gives rise to immense phenomenological richness: it concurrently supports volumetric pressure, shear stress, phase, topology, vortical circulation, memory, and resonance. This reflects Philip Anderson's celebrated insight—**"More is Different"**: every emergent level of structural complexity exhibits its own autonomous physical laws [Anderson, 1972]. RefG understands transitions across scales as the self-organization of a multi-channel medium.

RefG characterizes the observable Universe as a **finite, boundaryless, self-measuring global resonator**. The foundation itself is pre-spatial and possesses no dimensional bounds; what is finite and boundaryless is the resonator generated by its self-distinction. The Universe measures itself via its own internal normal modes.

6.2 Three Operational Infinities

Measurement of the Universe is performed exclusively using instruments that are themselves internal components of the system: our clocks, measuring rods, detectors, and physical bodies are oscillon configurations of the universal medium. This concept of a **"self-measuring Universe"** resonates with John Wheeler's *It from Bit* program [Wheeler, 1990]; in RefG, however, the physical foundation-medium is primary, and information is the measurable trace of its self-distinction.

Here, "infinity" carries a strictly **operational meaning**: it designates the asymptotic boundary of a measurement procedure, without asserting actual physical infinities in nature.

Kurt Gödel's incompleteness theorems [Gödel, 1931] provide an epistemological analogue: just as a consistent formal mathematical system contains true statements that cannot be proven from within the system, a self-measuring Universe raises a fundamental question: *To what extent can an internal observer measure the global boundary conditions of the entire system?*

This operational constraint gives rise to **"Three Operational Infinities"**:

1. **The Operational Infinity of Time:** Prior to the formation of stable resonant oscillons (clocks), operational time was undefined. The value "13.8 billion years" represents metric time calibrated against present-day atomic standards. Process time describes the early Universe as possessing a larger internal cycle budget, while metric cosmological time preserves its 13.8 Gyr empirical calibration.

2. **The Operational Infinity of Space:** We cannot measure abstract "empty space"—we measure only wave coupling between distinguishable oscillon nodes. Consequently, the cosmological horizon marks the operational boundary of our causal measurement network rather than a physical termination of the foundation-medium.

3. **The Operational Infinity of Scale:** In RefG, the Planck length ($\ell_{\mathrm{Pl}} \sim 10^{-35}$ m) marks not an absolute geometric continuum floor, but the natural transition scale constructed from c, $\hbar$, and G where quantum and gravitational effects achieve parity. Below this scale may lie a deeper dynamical stratum of the foundation, or the concept of spatial extension may cease to apply.

In each case, infinity emerges from the divergence between an operational measurement procedure and the demand for exhaustive description. A finite domain can project as infinite under specific coordinate transformations (analogous to the stereographic projection of a sphere onto an infinite plane or Benoit Mandelbrot's coastline paradox; Mandelbrot 1982).

6.3 Methodological Principles

The methodological foundation of this research program is absolute: **An intuitive physical picture must be translated into a rigorous, computable, mathematically self-consistent, and falsifiable formalism.**

The theoretical analysis of a particle oscillon follows a rigorous 9-step verification chain:

1. **Empirical Grounding:** Precise determination of mass, charge, spin, lifetime, and scattering form factors;
2. **Mode Identification:** Selecting the appropriate resonant and topological class within the foundation-medium;
3. **Geometric Ansatz:** Formulating the spatial geometry, topological invariants (Lk, Wr, Tw), and boundary conditions;
4. **Dynamical Law:** Deriving the nonlinear field action, Lagrangian density, and equations of motion;
5. **Symmetries and Conservation Laws:** Identifying Noether currents, charge, spin, color, and zero modes;
6. **Existence and Stability:** Proving the existence of localized finite-energy soliton solutions and analyzing their perturbation spectrum;
7. **Parameter Closure:** Fixing all free coefficients independently of the data to be predicted;
8. **Novel Predictions:** Computing independent, measurable physical observables and their uncertainties;
9. **Empirical Confrontation:** Testing predictions against observational data under pre-defined falsification criteria.

A working hypothesis and a verified computational result are strictly distinguished. Every calculation is tied to reproducible code, explicit datasets, and executable verification gates.

6.4 Where This Program Could Die (The Falsification Map)

The hallmark of a legitimate scientific theory is its vulnerability to empirical refutation (Popper's criterion). RefG establishes a transparent, three-tiered falsification map:

1. The Entire Framework is Falsified if:

1. The oscillon sector fails to recover effective Lorentz invariance and spatial isotropy of light speed ($|\Delta c/c| > 10^{-17}$; Herrmann et al. 2009), or gravitational-wave propagation speed deviates from c ($|v_{\rm gw} - c|/c > 10^{-15}$; Abbott et al. 2017).
2. The universality of free fall between inertial and gravitational mass exhibits a composition-dependent violation exceeding observational bounds ($\eta > 10^{-15}$; MICROSCOPE: Touboul et al. 2022).
3. The global pressure evolution of the foundation-medium fails to yield a unique relation between process and metric time, or the cosmology built from that relation fails to recover the observed CMB, BBN, photon redshift, and Type Ia supernova $(1+z)$ time-dilation relations [Blondin et al., 2008].
4. Dimensionless coupling constants vary under local gravitational potential shifts ($|\Delta\alpha/\alpha| > 10^{-6}$; Kotuš et al. 2017).
5. Self-distinction of the foundation fails to establish a unified bridge to mutually distinguishable modes, local causality, and additive accounting of energy, momentum, and charge.

2. An Individual Sector is Falsified if:

1. Gravitational-wave energy flux fails to reproduce binary pulsar orbital decay to a precision of 10^{-3}–10^{-4} [Kramer et al., 2021], or scalar breathing modes exceed LIGO-Virgo-KAGRA bounds.

2. The galactic vortex framework fails to reproduce SPARC/RAR data [McGaugh et al., 2016] or the 8σ separation between lensing centroids and X-ray gas in the Bullet Cluster [Clowe et al., 2006].

3. Combined vortical-nodal sources fail to recover primary CMB anisotropies, the BAO $100\,h^{-1}$ Mpc ruler, and large-scale structure statistics [Eisenstein et al., 2005].

4. The compact regular-core model fails to match Event Horizon Telescope shadow diameters, the compact-branch oscillation spectrum, or gravitational-wave echo limits [Cardoso and Pani, 2019].

3. A Specific Geometric Reading is Falsified if:

1. The electron oscillon profile fails to maintain a pointlike electromagnetic form factor down to 10^{-18} m [Navas et al., 2024].

2. The Koide/C_3 branch fails to construct an action-derived radiative bridge to pole masses, or a fourth sequential charged lepton is discovered.

3. Galactic-flow circulation is not quantized in units of h/m within an independently fixed effective-mass and coherence window.

4. The Möbius double-cover fails to yield the Dirac value $g = 2$, Fock-space antisymmetry, or QED radiative corrections.

Each falsification criterion is tied to specific experimental measurements: an adverse result eliminates the corresponding hypothesis from the research program rather than prompting ad-hoc verbal revisions.

Conclusion

This monograph has established the unified conceptual architecture of **Refractive Gravity** (RefG):

1. **Ontological Grounding:** The spontaneous self-distinction of a single universal, pre-spatial, and pre-temporal foundation generates the operational origins of measurability—yielding space, time, and mass.

2. **Matter as Oscillons:** Locally confined, topologically stable nonlinear wave configurations of the foundation—**oscillons**—induce a static pressure deficit in the surrounding substrate via Bernoulli-type dynamics ($-\Delta P \propto \rho_{\mathrm{dyn}}$), registered by external observers as gravitational mass.

3. **Gravity as Refraction:** The propagation of matter and light wavepackets through an inhomogeneous pressure landscape constitutes universal optical-geometric refraction; along the static weak-field 1PN branch, this mechanism recovers the predictions of General Relativity (PPN $\beta = \gamma = 1$).

This unified pressure chain unfolds across three primary astrophysical scales:

- **Locally:** In compact relativistic objects, where internal topological content and externally read mass decouple via two-channel accounting;

- **Galactically:** In the coherent vortical circulation and polarization of the foundation, which RefG connects to MOND and Tully–Fisher (RAR/BTFR) regularities without dark matter particles;

- **Cosmologically:** In the global thermodynamic relaxation of the foundation-medium, wherein the cosmological constant Λ is interpreted as residual vacuum stress.

Concurrently, the framework connects material discreteness to the eigenspectrum of a 3D volumetric resonator and population-locked synchronization; C_3 generation symmetry and the ninth-order cycle $(\theta = 2/9, A_K = \sqrt{2})$ provide a geometric map of Koide's lepton mass formula; and the topological invariants $Lk = Wr + Tw$ organize spin, charge, and color quantum numbers within one geometric language.

Refractive Gravity constructs a comprehensive research program aimed at deriving a unified, computable, and empirically testable account of gravitation, matter, and measurability from a single ontological foundation. This monograph establishes a firm foundation for that physical paradigm and opens a broad path for further theoretical and experimental research.

Current Status of the Research Program

The current status of each theoretical component is cataloged here to maintain clarity and transparency across the research program:

1. **Ontological Foundation and the Origin of Measurability**—The hypothesis of a single pre-spatial and pre-temporal primary identity has been formulated. Its resonant self-distinction generates distinguishable nodes, quantity, operational spacetime, material multiplicity, and local causality. A complete dynamical derivation of this transition remains an active goal.

2. **Inter-Node Separation and Minimum Distinguishability**—The operational concept of distinguishability has been established; a Planck-scale cutoff serves as a physical candidate, though its exact derivation from the underlying action remains open.

3. **Local Collective Cadence**—Common-rescaling invariance and trial models of mutual synchronization have been obtained; calculating the complete spectrum of stable oscillons remains an active task.

4. **Oscillon Deficit, Scale, Time, and Mass**—Metric filters and the first-order biconformal branch have been derived symbolically; a finite-energy nonlinear oscillon profile and the full mass-readout map remain open.

5. **Weak-Field Gravity and EFT Architecture**—On the pressure-deficit channel, $H = 0$, $\gamma = 1$, and $\beta = 1$ are obtained. From conservation of oscillon topological-defect density, macroscopic four-dimensional covariance, and the effective postulates of Fierz–Pauli dynamics, Lovelock's and Deser's theorems yield a conditional low-energy General-Relativistic architecture within effective field theory. A direct nonlinear derivation from the microdynamics remains open.

6. **Inertia and Gravitational Mass**—The Noether identity is conditionally established for finite-energy, Lorentz-covariant localized solutions satisfying the Laue boundary condition; an exact derivation of the zero mode remains open.

7. **Gravitational Waves and Local Unobservability**—In the local Minkowski limit, tensor propagation speed, two transverse-traceless (TT) polarizations arising from shear modes, and detector response are established; cosmological propagation, source-flux normalization, the full waveform, and scalar suppression bounds remain open.

8. **Compact Objects**—A static, conditional map and finite-core matching of the C^2 ansatz have been obtained; within one specific exterior-layer and pressure-reversal criterion, admissible domains do not overlap. The physical equation of state, gravitational collapse, rotation, and global stability remain open.

9. **The Galactic Vortex and MOND/Tully–Fisher**—The macroscopic polarization W-potential yields vortex closure, the simple interpolation function, and asymptotic BTFR ($v^4 = GM_b a_0$). With fixed a_0 and mass-to-light ratios, the selected 163-galaxy, 2,788-point SPARC comparison gives an RMS scatter of 0.141 dex and a mean residual of -0.012 dex. In one Miyamoto–Nagai configuration, a 300×300-grid QUMOND estimate gives a maximum radial correction of 9.23%. On the ideal logarithmic branch, the lensing angle matches the singular-isothermal-sphere result. The microscopic origins of W and a_0, a full SPARC likelihood analysis, grid and boundary convergence of the disk solution, general geometries, and complete cluster dynamics remain open.

10. **Cosmological Relaxation, the Time Bridge, and Dark Energy**—Within the pressure–phase postulate and the selected $P(a)$ branch, an exact coordinate bridge from process to metric time has been obtained, $d\tau/dt = a^{-3/4}$. The case $\alpha = -1/2$ is the special branch of constant vacuum relaxation in process time. A joint fit of the free-α branch to 31 cosmic-chronometer measurements and 40 Pantheon bins, with full covariance, gives $\chi^2_{\text{tot}} = 53.75$ (ΛCDM: 53.94), $\Delta AIC = +1.81$, $\Delta BIC = +4.07$, a metric age $t_0 = 13.62$ Gyr, and a present-clock-calibrated process interval $\tau_0 = 29.86$ Gyr. Unchanged extrapolation of the unsaturated branch enters a phantom regime, $q_\infty \approx -2.01$. The next computational stages are linear cosmological perturbations and structure growth, a full-covariance BAO test, BBN, the full Planck CMB angular anisotropy spectrum, and the late-time relaxation law.

11. **Two-Level Time and Quantum Randomness**—The interpretive framework is compatible with Bell tests and delayed-choice experiments; formal derivations of Born's rule, the no-signaling condition, and measurement dynamics remain open.

12. **Particles as Oscillon Modes**—The geometric placement map and initial spectral classification have been formulated as candidates; finite-energy oscillons, form factors, masses, couplings, and decay channels remain open.

13. **The Koide/C_3/2/9 Structure**—The C_3 algebra yields $K = 2/3$ exactly; with $(A_K, \theta) = (\sqrt{2}, 2/9)$ anchored to the electron pole mass, tree-level predictions deviate by 451.7σ for the muon and 0.61σ for the tau. The choice $h = 2$, $m \propto \nu^2$, action-level derivation, and radiative corrections remain open.

14. **Spin, Pauli Exclusion, and Fermionic Statistics**—The Möbius double-covered geometry provides a candidate mechanism; a complete derivation of Fock space and the Spin-Statistics Theorem from first principles remains open.

15. **Charge, the Photon, and the Electromagnetic Channel**—Local-frame $U(1)$ symmetry, two helicities, screening of the hedgehog's linear tail, the surviving asymptotic $1/r^2$ field and flux accounting, and conserved-phase-current subtheorems have been obtained; canonical charge quantization, Maxwell's equations, Gauss's law, Ward identities, QED, the Dirac sector, and the exact gravitational coupling of free radiation remain open.

16. **Color, Quarks, and Gluons**—The three-color, eight-mode construction provides only a partially compatible structural map; $SU(3)$ structure constants, interaction vertices, scale evolution, the confinement potential, and fractional charge normalization remain open.

17. **Electroweak Map and the Higgs**—This constitutes a geometric placement map; $SU(2)_L$ chirality, a W/Z mass matrix preserving a massless photon, CKM/PMNS mixing, neutrino mass hierarchy, Yukawa couplings, and the vacuum expectation value remain open.

18. **The Fine-Structure Constant α**—A nodal interpretation and target-independent spectral measure have been obtained: α is read from the relative strengths of the electric-flux and transverse-photon response channels, with the overall scale canceling. A microscopic action for physical charge Q, a unique numerical value, the complete charged spectrum, and RG flow to the

Thomson limit remain open. The earlier boundary formula remains a target-dependent conditional result.

19. **The Gravitational Hierarchy**—The force ratio is expressed algebraically as $F_{\rm em}/F_g = \alpha(M_{\rm Pl}/m)^2$; the fundamental origins of G, $M_{\rm Pl}$, the oscillon energy scale, and the mass hierarchy remain open.

20. **A Finite, Self-Measuring Manifested Universe**—The global volumetric resonator is a cosmological hypothesis; low-multipole and topological signatures serve as observational candidates, with unique likelihood matching to data pending further investigation.

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